

Query-Efficient Quantum Approximate Optimization via Graph-Conditioned Trust Regions

Molena Huynh
North Carolina State University
Molena.Huynh@jmp.com

Abstract

A good transferred QAOA solution can leave improvements that are difficult to propose and resolve with finite measurements. We study this regime for MaxCut by relating trust-region radius, affine-model bias, propagated shot variance, and acceptance power. Smaller regions improve exact-data proposals in the diagnostic sample, but amplify gradient noise and leave smaller decrease margins. A conditional bias–variance calculation predicts gradient mean-square error before independent observations: pooled error on eight fresh graphs is 1.5% above its simulator-informed prediction. A separate 256-run factorial isolates radius-dependent sampling from stopping after unresolved evidence. All four policies return identical vectors within each matched run; stopping reduces expenditure without a detected gain from adaptive sampling. Four-look endpoint replays find only 2.33% mean Bernstein acceptance in the fixed positive-margin cohort at the largest online endpoint-pair allowance. The preserved 256-run revision accepts four of 288 proposals, with shaped gains below the SPSA and COBYLA means. Restoring the original shaped comparator exposes a depth-dependent quality–cost tradeoff. These results quantify model accuracy and decision resolution after transfer. They establish neither an optimizer-complexity advantage nor a measurement-only controller: the predictive diagnostics use exact simulator quantities, and hardware validation remains open.

1 Introduction

The quantum approximate optimization algorithm (QAOA) searches a family of parameterized quantum states for a large expected objective value [11]. Its classical parameter dimension can be small at shallow depth, but each objective estimate requires repeated circuit measurements. Reusing parameters from a related graph offers an appealing way to reduce this cost. Once transfer produces a good starting point, however, the relevant question changes: can a finite measurement budget resolve the improvements that remain?

We examine this question for MaxCut through a finite-shot local-refinement policy with a graph-selected anchor. Graph comparison selects a donor vector; nearby, symmetry-aligned donors set a bounded search shape; an affine model proposes a move; and fresh endpoint measurements determine acceptance. This construction makes four possible sources of benefit separately testable. Descriptor selection is the default; global transport and local-neighborhood scores are controlled comparisons rather than presumed improvements. An objective-estimation call, or query, is one newly sampled circuit estimate at one angle vector. We report its shot cost and its grouping into logical batches, because fewer calls need not imply fewer circuit executions or lower total cost.

Parameter transfer has several established foundations. Depth-dependent patterns motivate interpolation and Fourier initialization [40], and discretized annealing provides structured initial schedules [30]. Concentration results motivate fixed-depth reuse within specified sparse random

graph ensembles [2]. Local subgraphs, degree parity, and learned graph representations support graph-dependent donor selection [10, 12]. Offline local-tree calculations and precomputed angles can also reduce online parameter search [34, 38]. More recent approaches generate graph-conditioned parameter trajectories [25] or refine a selected layer after donor transfer [27]. These approaches differ in training, preprocessing, and online measurement costs; their common motivation makes initialization-only controls essential when assessing additional refinement.

Graph information supplies an anchor and a search shape. The default is an inexpensive descriptor; transport and rooted-neighborhood comparators are supporting controls. Their role is to test the initialization premise, rather than account for improvements that require new measurements. Detailed structural constructions and their limitations are given in Methods.

Local modeling and shot allocation also have substantial precedent. QR-based least squares and interpolation geometry control numerical sensitivity [5, 14]. Sung et al. use local quadratic models and observation reuse for variational quantum optimization [35]; ANASTAARS builds noise-aware models in adaptive random subspaces [8]. The iCANS optimizer allocates shots to gradient components [21], while the QAOA method of Ha, Shashaani, and Menickelly models both the objective and its spatial variance [15]. Stochastic trust-region and adaptive-sampling convergence theories impose their own accuracy and sampling conditions [4, 31]. Our model is full-dimensional and affine, with a fixed-volume graph-derived shape. The revised policy allocates model shots predictably from the radius, reserves a first endpoint test, and stops when the declared decision schedule remains unresolved. These choices differ from spatial objective-and-variance modeling and latency-aware two-stage estimation in Ref. [15]. Empirical-Bernstein acceptance alone is an established statistical ingredient, not a novelty claim. Our variance-sensitive comparator implements this endpoint rule; it is not a reproduction of that published trust-region method. A matched comparison with the latter remains necessary for a claim of superiority over variance-aware QAOA optimization.

The central scientific question is whether a measured relationship between radius, fitting bias, shot variance, and acceptance power predicts useful refinement conditions. We test this relationship in three stages: audit the saved decisions and restore the original shaped comparator; isolate sampling and stopping through a separate factorial experiment; then evaluate fixed predictions on fresh graph instances. The bias-variance identity and empirical-Bernstein inequality are standard tools. Their role here is to make a conditional QAOA prediction testable, with exact-data inputs and measurement-only decisions clearly distinguished. Matched quality-cost comparisons determine whether this construction improves query efficiency.

2 Results

2.1 Separating initialization from refinement

Figure 1 exposes the comparison that a refinement claim needs. Returning the selected anchor is a complete initialization-only policy; fitting and accepting a new point incur additional target-graph measurements. The objective is normalized by the number of edges, with $f_G = -F_G/|E| \in [-1, 0]$, so a decrease in f_G is an increase in expected cut. Two quantities separate the two stages. The reported deficit $D_G = 100[f_G(\theta_{\text{out}}) - f_G^{\text{ref}}]$ measures the returned vector against the best-found reference, so smaller is better and negative values are possible because that reference is not a certified optimum. The incremental improvement $I_G = 100[f_G(\theta_0) - f_G(\theta_{\text{out}})]$ measures the returned vector against the policy’s own initialization, so $I_G > 0$ exactly when refinement improved on the anchor it started from and $I_G = 0$ whenever the returned vector equals its anchor. Reporting both prevents a good anchor from being credited to the refinement algorithm.

2.2 Matched refinement and measurement trajectories

The historical experiments motivate the revision; they are preserved in Appendix A. The new experiment reuses the follow-up’s 16 target graphs, two fixed banks, depths two and three, and validation-selected settings. It is a post hoc controlled intervention on those instances, not an independent replication. The earlier revised four-cell design holds the anchor, shot cap, model family, and settings fixed while crossing isotropic or shaped regions with Hoeffding or empirical-Bernstein acceptance. It does not isolate sampling from stopping. The new attribution experiment holds shaped geometry and Bernstein acceptance fixed and crosses fixed or radius-dependent shots with stopping or continuing after unresolved evidence.

A matched revision separates transfer from refinement. The post-hoc revision evaluates 256 new runs on the same 16 follow-up target graphs, with byte-identical graph-specific descriptor anchors and the previously selected initial radius and model-shot count. The full isotropic/shaped by Hoeffding/empirical-Bernstein factorial changes the radius-dependent sampling and stopping logic without retuning on those targets. Its measured cost is 9,458,244 optimization shots plus 2,097,152 independent output-validation shots. The historical SPSSA, COBYLA and original Bernstein runs provide matched-anchor comparators at the same cap. The original Bernstein rule is our own implementation; it is not an implementation of a published variance-aware optimizer. The revision is exploratory because these graphs were already used to diagnose the original method.

Returned quality and incremental gain answer different questions. At depth 2, revised shaped Bernstein has paired refinement gain 0.07 (95% interval 0.00 to 0.20) percentage points and changes 2/32 outputs. The original shaped Bernstein gain is 0.032508 percentage points. Using unrounded paired graph records, the revision minus original gain difference is 0.032665 (95% interval 0.000000 to 0.097995) percentage points, and its mean shot difference is -24284.38 (95% interval -27360.00 to -20536.75). The number of unique targets changed is 1 for the revision and 1 for the original. These depth-specific paired comparisons separate gain from avoided expenditure. Its paired gain difference from SPSSA is -0.13 (95% interval -0.21 to -0.05) percentage points; positive values favor shaped Bernstein. Its paired gain difference from COBYLA is -0.02 (95% interval -0.15 to 0.09) percentage points; positive values favor shaped Bernstein. At depth 3, revised shaped Bernstein has paired refinement gain 0.03 (95% interval 0.00 to 0.10) percentage points and changes 1/32 outputs. The original shaped Bernstein gain is 0.108996 percentage points. Using unrounded paired graph records, the revision minus original gain difference is -0.077215 (95% interval -0.192421 to 0.000000) percentage points, and its mean shot difference is -26243.31 (95% interval -29216.00 to -22534.62). The number of unique targets changed is 1 for the revision and 2 for the original. These depth-specific paired comparisons separate gain from avoided expenditure. Its paired gain difference from SPSSA is -0.33 (95% interval -0.47 to -0.20) percentage points; positive values favor shaped Bernstein. Its paired gain difference from COBYLA is -0.16 (95% interval -0.37 to 0.00) percentage points; positive values favor shaped Bernstein. Figure 1 reports quality, own-anchor gain, actual shots and changed-output counts together. Figure 2 uses completed incumbent events rather than connecting separate budget-cap means: initialization is available at zero shots, and an unfinished atomic batch leaves the preceding output available. Target curves record first-ever attainment, retain failed runs in their denominator and count initial successes at zero cost. The paired shaped-policy difference uses pointwise graph-block bootstrap bands. Flat tails after termination mean that the output remains available without further expenditure. Changing a parameter vector does not guarantee objective improvement for an unconstrained baseline. The reference objective is best found, not a certified depth-matched optimum; signed deficits remain signed.

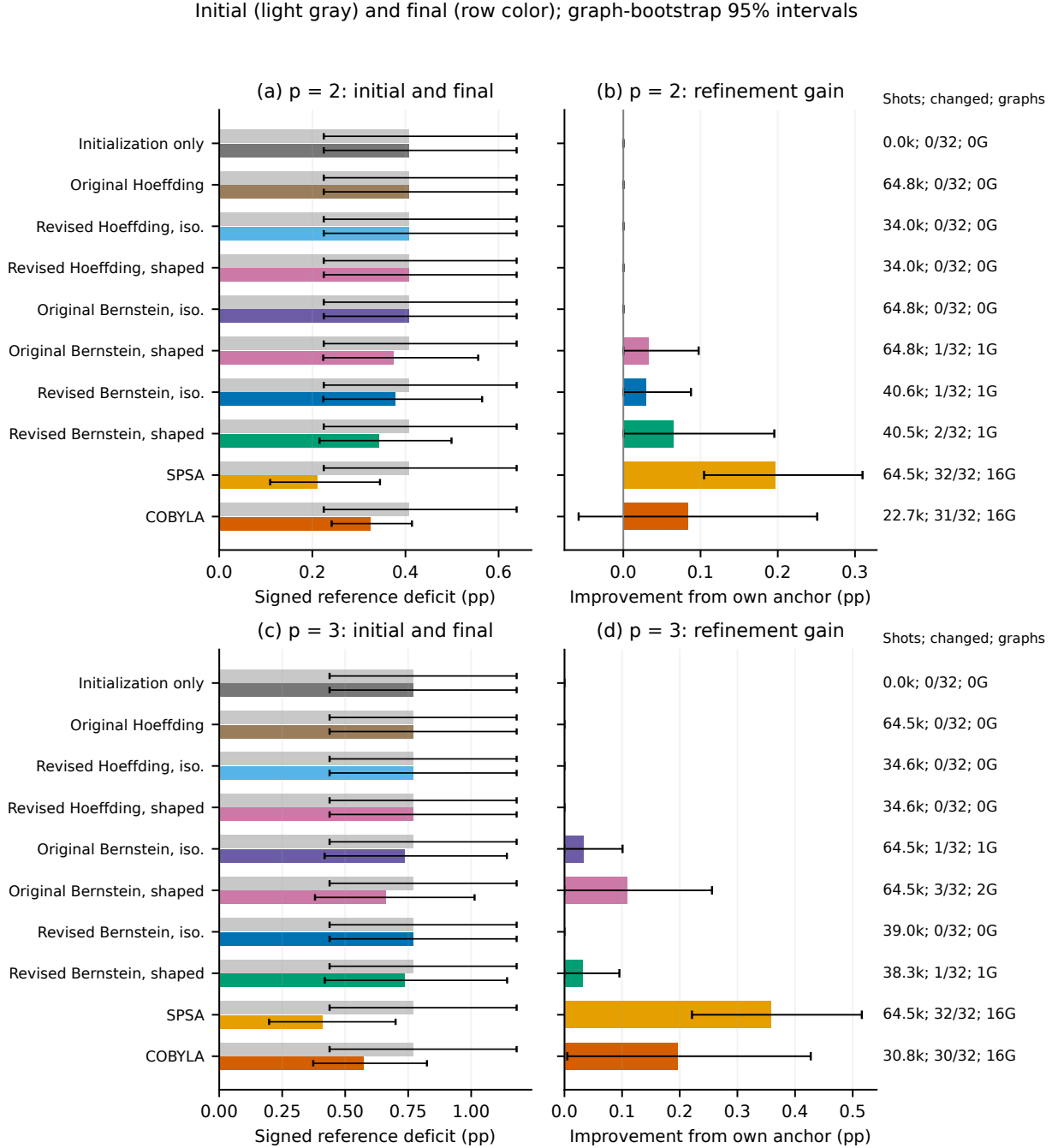


Figure 1: Initialization and actual refinement at depths two and three. Grouped bars compare signed initial and final reference deficits and gains relative to each policy’s own common descriptor anchor. Initialization-only has exactly zero gain. Revised policies and preserved matched baselines use a 65536-shot cap; annotations give actual optimization expenditure and changed outputs and unique graphs changed. A changed vector need not improve the true objective for an unconstrained baseline. Original shaped Bernstein is included as the direct historical comparator. Descriptive 95% intervals resample graph-level means within the two fixed panels, after averaging shot repetitions. A negative deficit is permitted because the reference is best found, not certified.

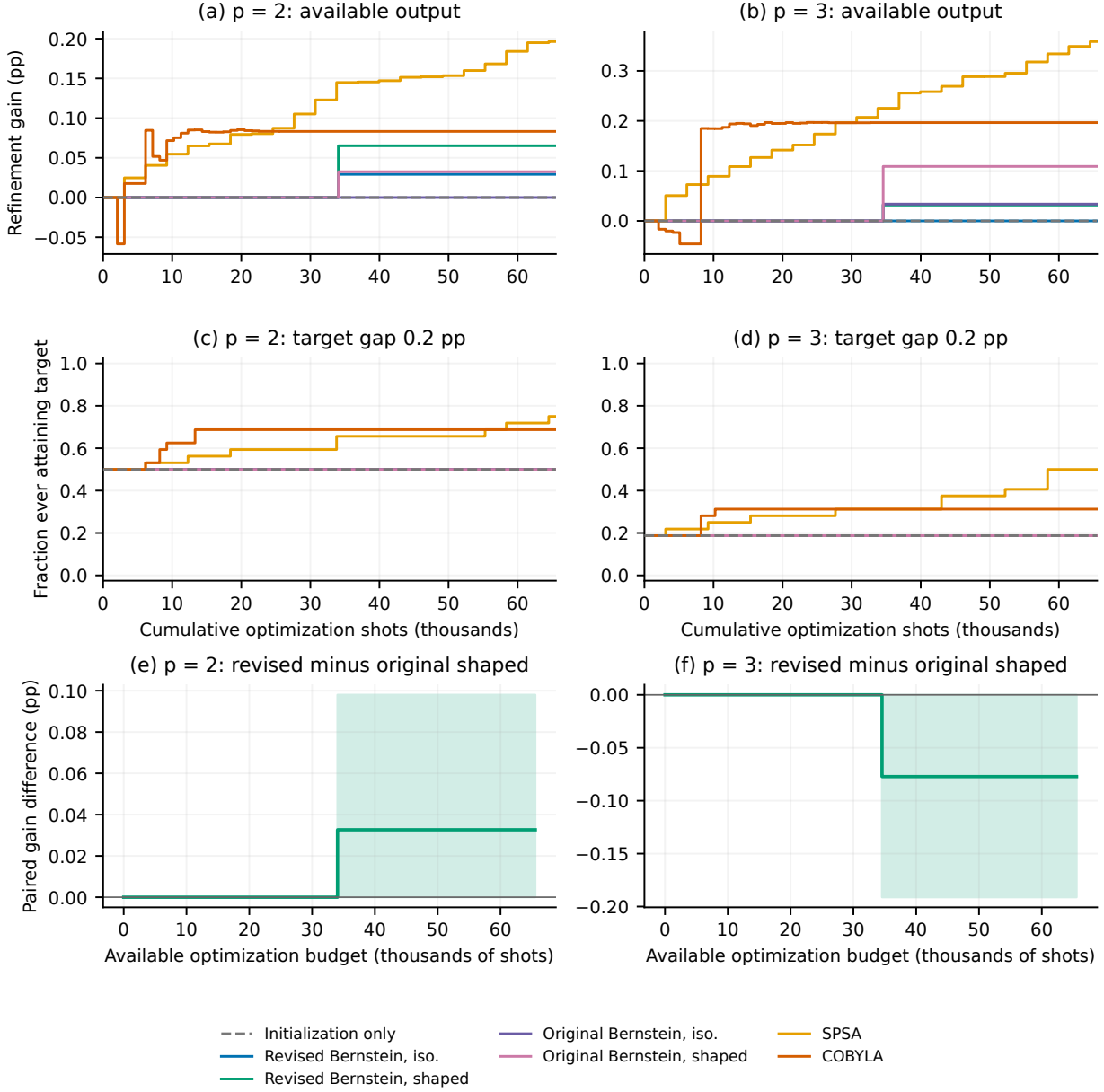


Figure 2: Available quality as measurement cost accumulates. Step lines use the last recorded incumbent available at each cumulative optimization-shot budget. An unfinished atomic batch leaves the preceding incumbent available. The zero-shot anchor is included. First-target-hit curves retain every run in the denominator and count initially qualifying instances at zero shots; the target gap is fixed before the revision measurements. The target is $f_G^{\text{ref}} + 0.002$. Exact expectations are retrospective diagnostics and never choose an output. These trajectories are extracted from runs at one cap, not interpolated between separate caps. Pointwise bands resample whole graph trajectories within the two fixed panels. The third row shows the paired gain difference between revised and original shaped Bernstein. Flat tails after termination mean that the saved output remains available, not that further shots are spent. First-ever success need not imply that a later nonmonotone baseline output still meets the target.

Table 1: Matched revision and historical comparators. Values are graph-mean refinement gains and signed reference deficits in percentage points, followed by actual mean optimization shots in thousands. All use the same descriptor anchor per graph. Add 8,192 independent validation shots per output. H and EB denote Hoeffding and empirical Bernstein; original EB is our own comparator. Graph-paired, fixed-bank-stratified 95% bootstrap intervals appear in the figures and text.

Policy	Gain		Deficit		Shots (10^3)	
	$p = 2$	$p = 3$	$p = 2$	$p = 3$	$p = 2$	$p = 3$
Initialization only	0.000	0.000	0.407	0.768	0.000	0.000
Original H, isotropic	0.000	0.000	0.407	0.768	64.768	64.512
Original EB, isotropic	0.000	0.034	0.407	0.735	64.768	64.512
Original EB, shaped	0.033	0.109	0.375	0.659	64.768	64.512
Revised H, isotropic	0.000	0.000	0.407	0.768	34.048	34.560
Revised H, shaped	0.000	0.000	0.407	0.768	34.048	34.560
Revised EB, isotropic	0.029	0.000	0.378	0.768	40.626	38.976
Revised EB, shaped	0.065	0.032	0.342	0.736	40.484	38.269
SPSA	0.196	0.359	0.211	0.409	64.512	64.512
COBYLA	0.083	0.197	0.324	0.572	22.720	30.752

2.3 Proposal quality and decision resolution

Write $a = f(x) - f(x^+)$ for a proposal’s true improvement and $h = a - \eta q$ for its sufficient-decrease margin. A nonimproving proposal has $a \leq 0$; an improving proposal with $h < 0$ still fails the declared sufficient-decrease requirement. At $h = 0$ it meets the weak decrease threshold but has no positive detection margin. On the simultaneous confidence event, a proposal with $h > 0$ may be accepted or remain unresolved. The 288 saved revised proposals comprise 178 nonimproving points, 30 improving points without a positive sufficient-decrease margin, 76 positive-margin points left unresolved, and four valid accepted improvements. The margin magnitudes and matched-rule replay determine whether unresolved proposals were close to the rule’s resolvable regime. These are proposal counts, not 288 independent graph observations. These categories require exact diagnostics only after proposal selection; the optimizer sees sampled means and variances. Figure 3 freezes proposals and redraws endpoint observations with independent random streams, so its acceptance frequencies measure sampling power rather than reusing the observations that generated the proposals.

Stopping status distinguishes limited resolution from adverse evidence. Across the new factorial, terminal statuses are `resolution_limited`: 256. An unresolved comparison does not certify a bad direction. A resolution-limited stop reports that the declared model or decision policy cannot resolve another update within its remaining budget or precision limits; it does not establish stationarity or optimality. The recorded reasons are `acceptance_look_cap`: 230, `next_acceptance_look_unaffordable`: 26.

Online power uses its own four-look confidence allocation. Figure 3 holds fixed the 58 positive-margin proposals and the 102 nonpositive-margin proposals, with equal proposal weights at every budget. The online row uses four prescribed looks up to 16,384 shots per endpoint; the extended row uses five looks up to 65,536. Both are recomputed from the same stored independent streams, so this comparison needs no additional sampled observations. Shading gives conservative pointwise 95% Hoeffding bounds for each fixed-cohort replay mean, allowing different endpoint dis-

tributions across proposals. Those intervals quantify Monte Carlo replay uncertainty conditional on the fixed proposals, not uncertainty over target graphs. At the 32,768-shot pair cap, online four-look Hoeffding has mean positive-margin acceptance frequency 0.0000 (pointwise interval 0.0000 to 0.0158), unresolved fraction 1.0000, and mean actual endpoint expenditure 32,768.00 shots. At the 32,768-shot pair cap, online four-look Bernstein has mean positive-margin acceptance frequency 0.0233 (pointwise interval 0.0075 to 0.0391), unresolved fraction 0.9767, and mean actual endpoint expenditure 32,759.17 shots. At the 131,072-shot pair cap, extended five-look Hoeffding has mean positive-margin acceptance frequency 0.0000 (pointwise interval 0.0000 to 0.0158), unresolved fraction 1.0000, and mean actual endpoint expenditure 131,072.00 shots. At the 131,072-shot pair cap, extended five-look Bernstein has mean positive-margin acceptance frequency 0.2877 (pointwise interval 0.2720 to 0.3035), unresolved fraction 0.7123, and mean actual endpoint expenditure 128,931.31 shots. For one pair, zero acceptances in 128 independent replays has exact one-sided 95% binomial upper limit 0.02313 (2.313%). Zero observed frequency therefore does not establish zero probability or empirically validate an analytical probability of order 10^{-6} .

Most original savings occur before radius-dependent allocation is activated. The preserved revision ledgers show that 31/256 runs fit a model away from the initial radius (32/288 fits), and 28/256 runs allocate more than 256 shots per point (28/288 fits). Every revised Hoeffding run fits one initial model and performs one full four-look endpoint comparison before unresolved stopping. Its costs are therefore exactly $5(256) + 2(16384) = 34048$ at depth two and $7(256) + 2(16384) = 34560$ at depth three. Those savings demonstrate avoided continuation, not an observed improvement from changing the model-shot allocation on the Hoeffding trajectories.

The implemented schedule is tested against its variance prediction. The mechanism study uses 8 original diagnostic graphs from panel zero and 8 newly generated targets nonisomorphic to all previous bank, validation and test graphs. Figure 4 displays shaped models on the original eight, with depths two and three equally averaged within each graph before bootstrap resampling. Full graph identities and their split membership are recorded in the saved protocol. Central-difference gradients at steps 10^{-5} and 5×10^{-6} differ by at most 1.39×10^{-10} in Euclidean norm across all targets and depths. This check supports the numerical reference used in the bias comparison; it does not turn a numerical derivative into an analytic exact-gradient formula. The 1920 target/depth/shape/radius/shot-count conditions each use 128 independent sampled fits. Root-mean-square error means the square root after averaging squared Euclidean errors, including graph averaging; it is not the arithmetic mean of error norms. Exact-data bias floors and the propagated covariance predict total mean-square error. The actual schedule samples 4096, 1024, 256, 64 and 16 shots per point at radii 0.025, 0.05, 0.1, 0.2 and 0.4. Its cost is multiplied by $2p + 1$ fitting points. Graph-level bands are pointwise 95% intervals conditional on the fixed bank and sampled designs; they do not provide simultaneous radius-wise coverage. Across the fresh-graph model grid, the ratios of mean observed to mean predicted sampling and total MSE are 1.016122 and 1.014871, respectively. On the implemented schedule alone, the corresponding ratios are 1.009032 and 1.002424. Both summaries equally average the fresh targets, depths and shapes. The identity predicts this second moment for the fixed fitting map; Gaussian positive-margin prediction additionally approximates the distribution of fitted objective means and the nonlinear proposal map.

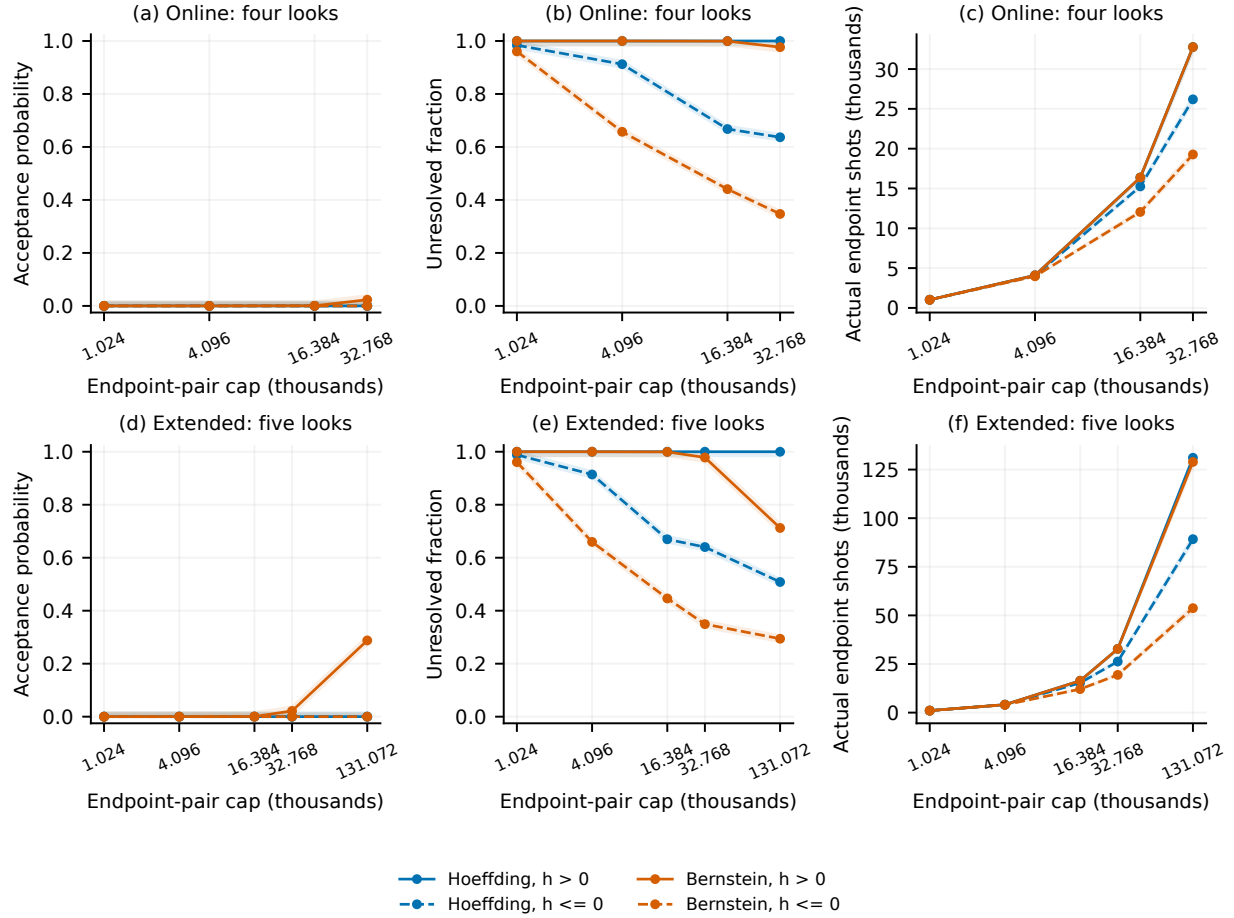


Figure 3: Acceptance power versus available endpoint-pair budget, with the same 160 frozen diagnostic proposals, fitted at 1024 shots per point, and equal proposal weights within each cohort at every budget point. This is conditional replay power, not the acceptance frequency of the optimizer runs. The first row uses the online four-look schedule and its $L = 4$ confidence penalty; the second uses the separately labelled five-look extension. Panels report acceptance probability, unresolved fraction, and actual endpoint expenditure, retaining unresolved cases at their last affordable look. Lines distinguish Hoeffding and Bernstein within fixed positive-margin and nonpositive-margin cohorts. Bands describe pointwise replay uncertainty conditional on these proposals, not generalization to other graphs. There are 128 replays per proposal; zero acceptances for one proposal give a one-sided 95% binomial upper limit of $1 - 0.05^{1/128} \simeq 2.31\%$, not proof of zero power or confirmation of a probability near 10^{-6} .

Shaped models: eight diagnostic graphs, depths 2 and 3 equally averaged

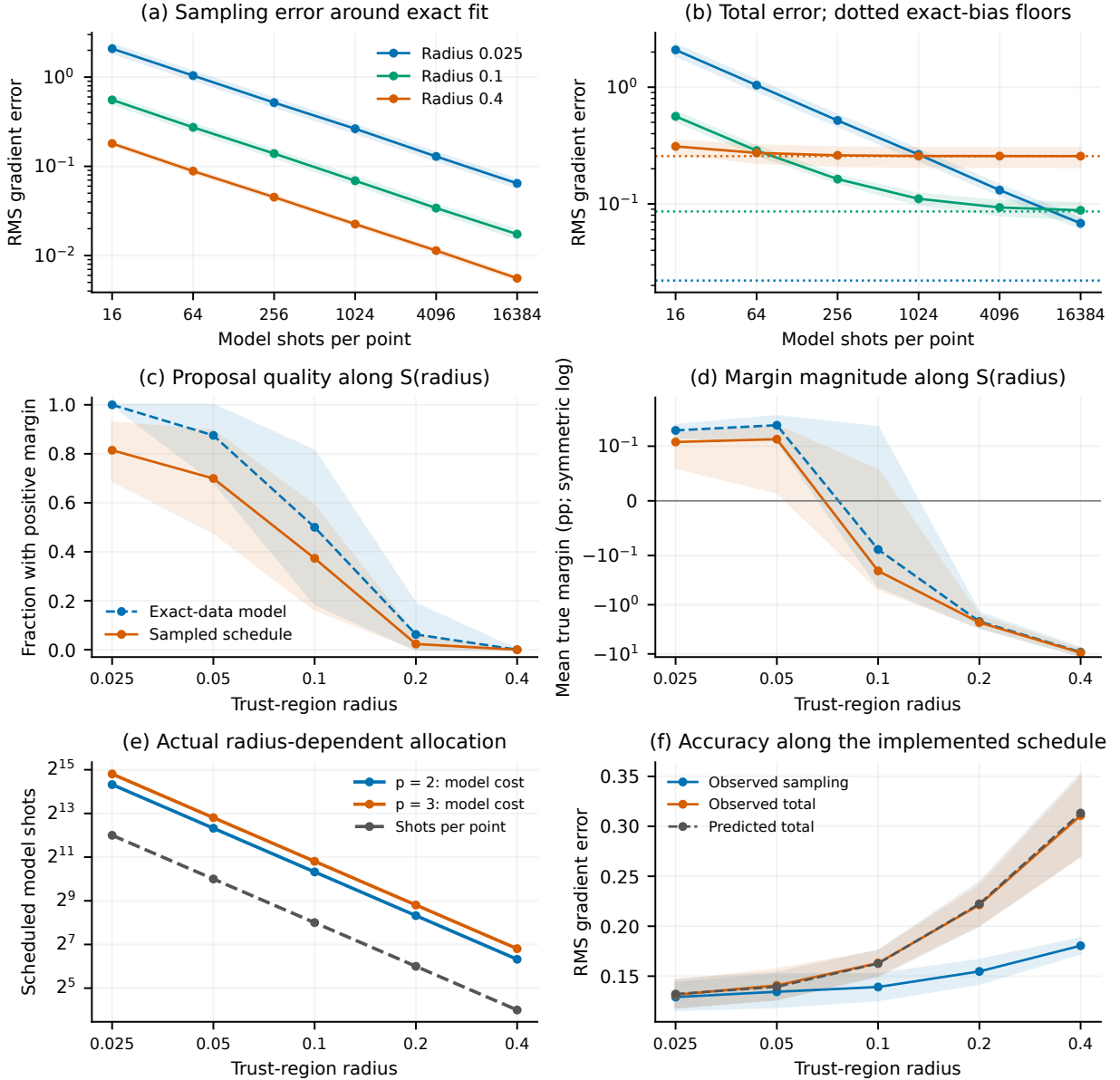


Figure 4: Radius, model bias, and propagated shot noise. Lines report RMS sampling error about an exact-data affine gradient, RMS total error against a numerically checked exact-objective gradient, and the exact-data bias floor. Radius panels show both positive-margin frequency and true margin magnitude; the margin axis is symmetric logarithmic with a linear region around zero. The schedule panels evaluate $S(\Delta)$ itself, including 64 and 16 shots at radii 0.2 and 0.4, and distinguish model expenditure from error. The diagnostic population is the first panel's eight graphs. Depths two and three are aggregated with equal graph weights as stated in the panels; pointwise bands resample graphs rather than treating repeated fits as new graphs. Predicted errors use the conditional variance-propagation identity; they are simulator-informed diagnostics rather than online oracle access.

Table 2: Controlled sampling-by-stopping attribution. Gains are percentage points over the common anchor; costs are actual optimization shots in thousands. Every row uses shaped empirical-Bernstein refinement at a 65,536-shot cap, with 8,192 independent validation shots per output. Two shot repetitions are averaged within each of 16 target graphs.

Allocation / unresolved policy	Gain (pp)		Shots (10^3)	
	$p = 2$	$p = 3$	$p = 2$	$p = 3$
Fixed / continue	0.0365	0.0723	64.7680	64.5120
Fixed / stop	0.0365	0.0723	39.0160	37.5840
Radius / continue	0.0365	0.0723	64.7752	63.8771
Radius / stop	0.0365	0.0723	39.9538	38.0696

2.4 Attribution and fresh-graph prediction

The interventions in Fig. 5 are distinct from the earlier shape-by-rule comparison. All four policies use the same anchor, shaped region, Bernstein test, cap, and paired random streams. Fixed versus radius-dependent model shots is one factor. Stopping versus refitting after unresolved evidence is the other; continuation leaves the radius unchanged until acceptance or certified rejection. Thus this experiment does not bundle continued sampling with the historical unresolved contraction rule.

Fresh-graph predictions are fixed before their new model observations. They use exact circuit expectations and variances to predict conditional model error; an approximate Gaussian coefficient model additionally predicts the frequency of positive sufficient-decrease margins. Independent finite-shot fits evaluate these predictions. This distinguishes a test of the proposed mechanism from optimization on those fresh graphs: it is not a new held-out ranking of complete optimizers. The descriptive size/depth effects are retained in Fig. 11 as supporting evidence.

A separate factorial isolates sampling allocation from unresolved stopping. The controlled attribution adds 256 shaped empirical-Bernstein runs on the original 16 targets, crossing fixed/radius-dependent model shots with stop/continue after unresolved evidence. All four cells share anchors, shape, model family, error allocation, shot cap, initial settings and paired seeds. Continuation holds the radius and refits from fresh data; it contracts only after certified rejection. This control is not a replay of the historical rule that contracted after unresolved evidence. Figure 5 separates gain and expenditure for these controlled policies. All four cells return byte-identical parameter vectors in 64/64 matched graph–depth–seed cases. In this controlled sample, neither continuation nor radius-dependent allocation changes returned quality; stopping changes expenditure. At depth 2 under fixed sampling, stopping minus continuation has paired shot difference -25752.00 (95% interval -28512.00 to -22440.00). At depth 2 under radius sampling, stopping minus continuation has paired shot difference -24821.44 (95% interval -28181.44 to -20789.44). At depth 2 with unresolved stop, radius-dependent minus fixed allocation has gain difference 0.00 (95% interval 0.00 to 0.00) percentage points and mean shot difference 937.81. At depth 2 with unresolved continue, radius-dependent minus fixed allocation has gain difference 0.00 (95% interval 0.00 to 0.00) percentage points and mean shot difference 7.25. At depth 3 under fixed sampling, stopping minus continuation has paired shot difference -26928.00 (95% interval -29952.00 to -23152.00). At depth 3 under radius sampling, stopping minus continuation has paired shot difference -25807.50 (95% interval -29952.00 to -20926.00). At depth 3 with unresolved stop, radius-dependent minus fixed allocation has gain difference 0.00 (95% interval 0.00 to 0.00) percentage points and mean shot difference 485.62. At depth 3 with unresolved continue, radius-dependent minus fixed allocation

has gain difference 0.00 (95% interval 0.00 to 0.00) percentage points and mean shot difference -634.88.

Fresh-graph prediction is simulator-informed and conditional. All fitting maps, covariance predictions and Gaussian proposal predictions were frozen before independent multinomial observations. The fresh targets use the same fixed panel-zero donor bank and no fresh-target tuning. Figure 5 compares 256 Gaussian proposal draws per condition with 32 independent sampled proposals along the implemented radius schedule. The mean absolute difference in positive-margin probability across shaped fresh conditions is 2.017 percentage points (graph-bootstrap interval 1.450 to 2.524). This discrepancy includes Gaussian approximation error and finite replay noise. The predictor uses exact simulator objective means, variances and trial expectations. It is an oracle-informed mechanism test, not a measurement-implementable regime selector, fresh-graph optimizer benchmark, or hardware validation. The additional attribution consumes 13,201,786 optimization and 2,097,152 independent validation shots. The mechanism observations require 5,367,398,400 simulated finite-shot outcomes; the Gaussian diagnostic uses 40,960 draws and prediction/reference construction uses 43,872 exact simulator queries, plus 10,240 exact sampled-proposal diagnostic objective evaluations. Those offline diagnostic costs are not credited as optimizer savings.

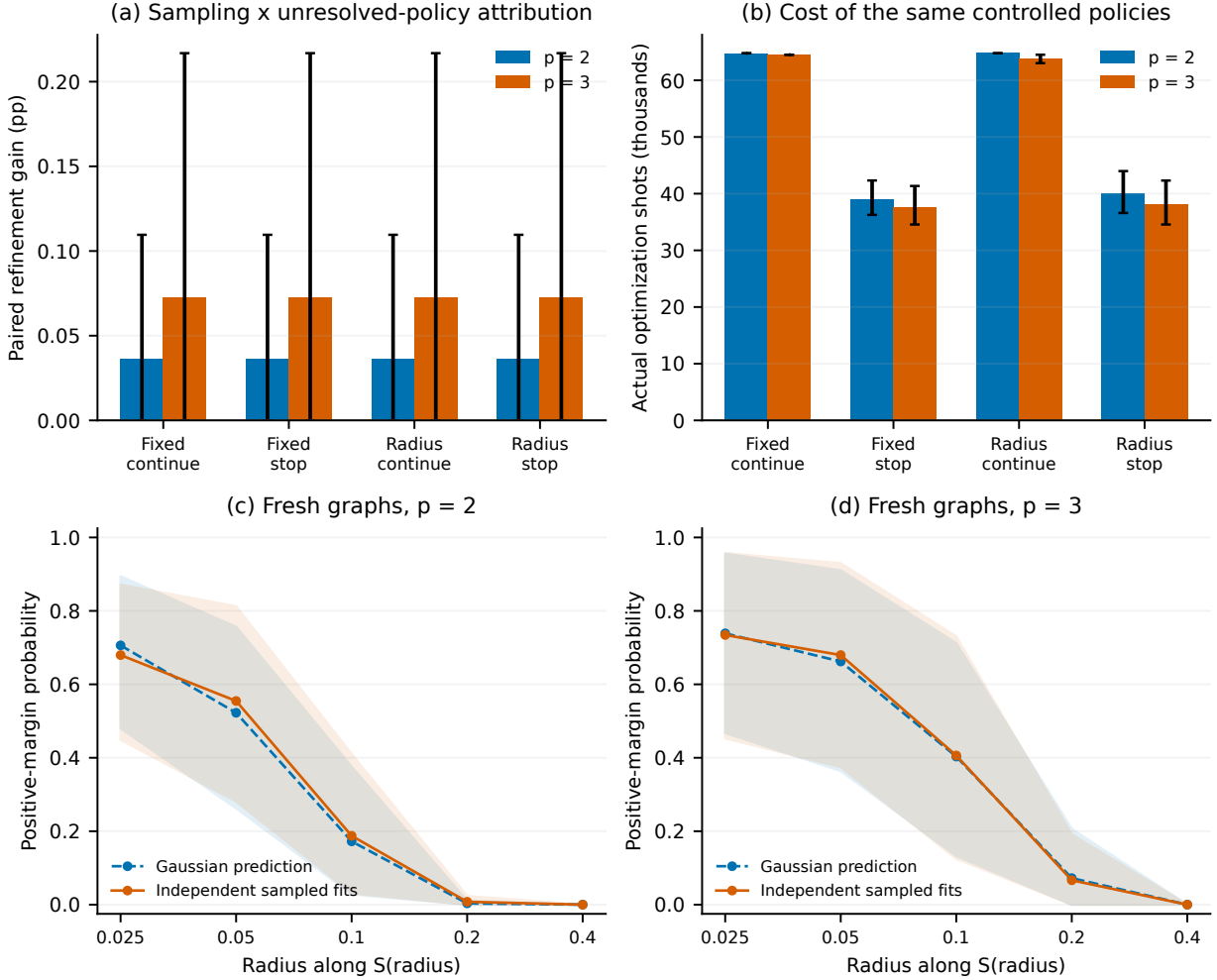


Figure 5: Which intervention changes refinement, and which prediction survives new targets? Grouped bars cross fixed or radius-dependent model shots with stop or continue after unresolved evidence, holding anchor, geometry, acceptance rule, and cap fixed. Separate panels report own-anchor gain and actual shots with paired graph-level uncertainty. Continuation refits at the same radius after an unresolved test. Lines on fresh graphs compare frozen simulator-informed predictions with independent observations across the prespecified radius range. These predictions test the mechanism; they do not assume a favorable outcome or demonstrate a measurement-only control rule.

3 Discussion

The central mechanism is a radius tradeoff. In the original diagnostic grid, all 32 graph/depth/shape combinations have a positive exact-data margin at radius 0.025, while none do at radius 0.4. Small regions can therefore repair the affine boundary proposal without making it easy to measure: the physical gradient fitting map amplifies sampling noise as the radius contracts, and a smaller move can leave a smaller improvement margin. Large-radius total error that persists with increasing shots instead points to model bias. Equation (18) separates these contributions and specifies a numerical prediction whose calibration can be tested on independent observations.

Three claims must be kept distinct. Earlier termination can avoid expenditure; a positive own-anchor gain establishes useful refinement; and a prediction fixed before new observations can establish a conditional model of the operating regime. None implies the others. The original shaped Bernstein policy belongs in the main comparison: its depth-three gain is about 0.109 percentage points, versus 0.032 for the first revision, whose mean expenditure is lower. At depth two the ordering of those gains reverses. These are quality–cost tradeoffs on a small graph sample, not a uniform improvement from the revised policy. SPSA retains larger mean gains at both depths, and COBYLA combines larger mean gains with fewer mean shots than revised shaping; graph-paired uncertainty accompanies these summaries.

The saved ledgers show where the sampling intervention actually operates. Only 31 of 256 revised runs fit a model at a changed radius, and 28 use more than 256 shots per point. Every revised Hoeffding run pays for exactly one initial model and one full four-look endpoint test. Its lower expenditure therefore supplies evidence about unresolved stopping, without testing higher model precision at a contracted radius. Moreover, 230 of the 256 terminal decisions reach the prescribed look cap and only 26 are unable to afford the next look. A policy-limited unresolved stop does not show that all potentially useful allocations of the remaining budget were exhausted. The separate sampling-by-stopping factorial makes this attribution direct: all four policies return identical vectors for each of the 64 matched graph/depth/repetition combinations. Their gains are 0.0365 and 0.0723 percentage points at depths two and three, while stopping reduces mean expenditure from roughly 64000–65000 to 38000–40000 shots. Changing model allocation alone yields no gain in this controlled sample. This conclusion concerns the specified hold-radius continuation policy, rather than every possible use of the remaining budget.

The four-look replay answers the online rule’s power question, conditional on the fixed proposal collection. It uses the online confidence penalty, not a prefix of the five-look experiment with its different error allocation. Fixing cohorts and weights across budget makes changes in power attributable to the allowed measurements rather than changing proposal distributions. Zero observed acceptance has finite Monte Carlo resolution: 128 zero-success replays of a single proposal give a one-sided 95% upper limit of about 2.31%. They cannot empirically validate the historical Hoeffding bound below 3.5×10^{-6} . That analytic bound remains a property of its specified rule, not a universal QAOA shot lower bound.

The predictive checks also have a precise scope. The gradient-error identity is standard linear-estimation mathematics; its exact means, variances, and reference derivatives are supplied by the ideal simulator. Fresh-graph calibration finds pooled total mean-square error 1.5% above the prediction on the eight new targets. This pooled agreement does not imply equal accuracy in every radius/shape/depth cell. The Gaussian coefficient model also tracks aggregate positive-margin frequencies across the tested radii, including the low-shot operating points, with finite Monte Carlo uncertainty. That separate nonlinear approximation remains an empirical hypothesis. Neither agreement supplies a measurement-only selector of useful radii or predicts whole-run efficiency. A deployable controller would need pilot-based estimates or justified bounds for its currently exact

inputs, would need to pay for them, and would require a separately frozen optimizer comparison.

Variance-aware QAOA optimization already includes spatial variance models and adaptive estimation [15]. Our comparisons of Hoeffding and Bernstein endpoint bounds do not reproduce that method, and the bias–variance identity does not replace such a benchmark. A strong algorithmic superiority claim would require that comparison and competent baseline tuning. The historical validation grid is tied, so its selected settings have no demonstrated tuning advantage. The continued policy used for attribution retains the radius after an unresolved test; it does not represent every possible continuation rule or reproduce historical continue-and-contract behavior.

Transfer is a separate premise. Descriptor anchors already meet the canonical target on all 24 graphs, and the nine canonical donor-based policies retain their anchors after 196,341,504 shots. The local certificate saturates on 301 of 384 depth-one pairs and every depth-two pair. Values below one are labelled not saturated, since practical usefulness does not follow from strict inequality. Deterministic ties remove a donor artifact without creating information in saturated scores. Transport results must likewise distinguish feasibility, donor changes, and quality on common valid-target sets; an aggregate audit time cannot support a per-solver runtime comparison.

The experiments use ideal sampling on small graphs, with no device noise, readout bias, drift, routing, or hardware latency. Exact donor construction, variance calculations, and reference gradients have classical costs. Logical batches and simulator times do not estimate hardware runtime. The finite horizon, positive radius floor, and absence of fully linear model accuracy preclude importing asymptotic trust-region convergence results. Time-uniform confidence sequences [19] are a possible alternative to finite looks, but their estimator, error allocation, and quality–cost performance would require separate validation.

The attribution experiment reuses existing targets; its shot repetitions add no independent graphs. The fresh eight-graph set evaluates a conditional mechanism, not a new optimizer ranking, structural-shift claim, or broad scaling law. Both old panels and the new targets use fixed banks; comparing banks on the same targets remains necessary to isolate bank variability. These limits are part of the result: query-efficient refinement requires a useful model, an affordable decision, and a favorable measured tradeoff. The present experiments test those requirements separately rather than inferring them from initialization quality or numerical conditioning alone.

4 Methods

4.1 QAOA objective, symmetries, and measurements

For an unweighted simple graph $G = (V, E)$ with $n = |V|$ vertices (one qubit each) and $m = |E| > 0$ edges, the cost and mixer Hamiltonians are

$$C_G = \sum_{(i,j) \in E} \frac{I - Z_i Z_j}{2}, \quad H_X = \sum_{i \in V} X_i. \quad (1)$$

With angles ordered as $\theta = (\gamma_1, \dots, \gamma_p, \beta_1, \dots, \beta_p)$, we use

$$|\psi_G(\theta)\rangle = e^{-i\beta_p H_X} e^{-i\gamma_p C_G} \dots e^{-i\beta_1 H_X} e^{-i\gamma_1 C_G} |+\rangle^{\otimes n}. \quad (2)$$

The minimization objective is $f_G(\theta) = -\langle \psi_G(\theta) | C_G | \psi_G(\theta) \rangle / m$. A computational-basis measurement z produces one observation $Y = -C_G(z) / m \in [-1, 0]$. Its mean is $f_G(\theta)$. The simulator samples the exact cut-value distribution by a multinomial draw; it does not replace the shot distribution by additive Gaussian noise.

Each γ has period 2π because the unweighted cut spectrum is integer. Each β has period $\pi/2$ in the objective because the associated global bit flip commutes with every cost layer and preserves the initial state up to phase. Simultaneously reversing all angles complex-conjugates the state and preserves the objective. Reference vectors are aligned to an anchor using these symmetries: for each of the two global sign choices, every coordinate is shifted by its period to the representative nearest the anchor, and the alignment with the smaller Euclidean distance to the anchor is retained. The procedure does not assume that every symmetry-equivalent optimum has been found.

All optimization decisions use shot estimates. Exact expectations are computed only for reference construction and recorded diagnostics, including post hoc objective trajectories. The reported expected approximation ratio is $F_G(\theta)/C_{\max}(G)$, with $C_{\max}$ obtained by exhaustive bitstring enumeration at these graph sizes. It is distinct from the largest cut observed among a collection of samples. A single depth- p circuit uses pm cost-edge rotations and pn mixer rotations; a two-CNOT implementation of each cost-edge rotation would use $2pm$ CNOTs per shot, before routing and compilation. We do not convert this logical count into a hardware runtime claim.

4.2 A structural certificate for objective transfer

For a graph $G = (V, E)$ with $m = |E| > 0$, let $F_G(\theta)$ be the expected MaxCut value at depth p , and define the minimized objective $f_G(\theta) = -F_G(\theta)/m \in [-1, 0]$. For an edge e , form the induced subgraph containing every vertex within distance p of either endpoint, and mark the endpoints as an unordered root pair. Write $\mu_{G,p}$ for the distribution of rooted isomorphism types obtained by sampling an edge uniformly from E .

The causal structure of QAOA yields the following uniform comparison.

Theorem 1 (Transfer through rooted edge neighborhoods). *For any two finite simple unweighted graphs with at least one edge, and any shared depth p , define $d_p(G, H) = \text{TV}(\mu_{G,p}, \mu_{H,p}) = \frac{1}{2} \sum_{\tau} |\mu_{G,p}(\tau) - \mu_{H,p}(\tau)|$, total variation in the convention that carries the factor one half, so that $d_p \in [0, 1]$ and $d_p = 1$ exactly when the two type distributions have disjoint supports. Then*

$$\sup_{\theta \in \mathbb{R}^{2p}} |f_G(\theta) - f_H(\theta)| \leq d_p(G, H). \quad (3)$$

If θ_H is ϵ -optimal for f_H on a common nonempty domain Ω , then

$$f_G(\theta_H) - \inf_{\theta \in \Omega} f_G(\theta) \leq \epsilon + 2d_p(G, H). \quad (4)$$

The proof below uses the established locality of fixed-depth QAOA edge expectations [2, 11]; related local-subgraph comparisons motivate parameter-transfer heuristics [12]. The formulation here makes the finite-graph total-variation and donor-regret consequences explicit. It gives a sufficient transfer certificate without requiring equal graph sizes or a vertex correspondence. It is conservative: different neighborhoods can produce the same edge expectation, and disjoint type distributions give $d_p = 1$, for which Eq. (3) gives no information beyond $f_G \in [-1, 0]$. The factor-two regret bound can be clipped at one. Equation (4) is additionally conditional on the donor: it presumes a value of ϵ for which θ_H is ϵ -optimal for f_H on Ω . The donor angles used here are the lowest objective among six BFGS runs on the exact expectation, a multistart local minimization that certifies no optimality gap, so no numerical ϵ is available and Eq. (4) is never evaluated for a graph pair in the empirical sections. Computing exact rooted isomorphism classes can also be expensive.

The global shortest-path score used in the GW construction does not appear in Eq. (3). The two objects are not measures on a common space: the GW coupling compares global shortest-path

relations under uniform *vertex* masses $a_i = 1/|V_G|$, whereas d_p compares rooted local neighborhood types under uniform *edge* sampling, and no theorem here connects a small GW distortion to a small d_p . Small GW distortion is therefore a heuristic reason to try transfer, not a certificate of objective agreement. Even a structural comparator with an exact certificate can be uninformative on a particular finite sample. The empirical comparison evaluates both the quality of the selected reference and the informativeness of the scores.

Proof of Theorem 1. For a fixed edge $e = (u, v)$, let $C_e = (I - Z_u Z_v)/2$. Propagate C_e backwards through Eq. (2). A mixer layer acts separately on qubits and cannot enlarge its support. Within a cost layer all ZZ gates commute. Gates with neither endpoint in the current support commute with the operator and cancel, while the remaining gates can add only immediate neighbors. Thus each cost layer increases the maximum graph distance from the original edge by at most one. After p layers, only the induced radius- p edge neighborhood can affect the expectation. Extra edges entirely on its outer boundary cause no problem: they are present in the induced graph, and gates that never meet the backwards support cancel.

The product initial state and the uniform circuit angles imply that root-preserving isomorphic neighborhoods have identical edge expectations. Hence, for a rooted type τ , there is a common function $h_p(\tau, \theta) \in [0, 1]$ such that

$$-f_G(\theta) = \sum_{\tau} \mu_{G,p}(\tau) h_p(\tau, \theta). \quad (5)$$

For any function taking values in $[0, 1]$, the difference of its expectations under two distributions is at most their total variation. This proves Eq. (3), uniformly in θ . Furthermore, $f_G(\theta_H) \leq f_H(\theta_H) + d_p$ and $\inf_{\Omega} f_H \leq \inf_{\Omega} f_G + d_p$. Combining these inequalities with the assumed ϵ -optimality proves Eq. (4). $\square$

The implementation counts exact rooted isomorphism classes. Degree and size invariants are used only to reject impossible matches, followed by an exact graph-isomorphism test with endpoint marks. No hash equality is treated as an isomorphism certificate.

4.3 Structural comparison and reference selection

For connected graphs, define $D_G(i, k) = \text{dist}_G(i, k)/\text{diam}(G)$ and uniform vertex weights $a_i = 1/|V_G|$, $b_j = 1/|V_H|$. A feasible coupling satisfies $T \geq 0$, $T\mathbf{1} = a$, and $T^\top \mathbf{1} = b$. Its squared structural distortion [24, 28] is

$$\mathcal{D}_{G,H}(T) = \sum_{i,k,j,\ell} (D_G(i, k) - D_H(j, \ell))^2 T_{ij} T_{k\ell}. \quad (6)$$

Following the regularized structural-comparison idea in Refs. [7, 28], the independent implementation starts from $T = ab^\top$ and applies damped entropic transport updates with regularization 0.05. With $K_{ij} = (D_G^2 a)_i + (D_H^2 b)_j - 2(D_G T D_H^\top)_{ij}$, the next proposal is obtained by Sinkhorn scaling the kernel $\exp(-2K/0.05)$, following the fixed-cost entropic transport construction [6]. The actual update averages the proposal and previous coupling with equal weights. The score reported is the *unregularized* distortion of the returned coupling.

The outer loop stops at a coupling-change Frobenius norm below 10^{-9} or after 100 updates. Log-domain inner scaling uses at most 500 iterations and a row-marginal tolerance of 10^{-12} ; both marginals are checked to 10^{-8} before accepting a returned coupling. The tensor K is rounded to 12 decimal places before forming the kernel. This fixed numerical guard suppresses roundoff-scale

symmetry breaking at the uniform start. It does not certify a stationary point, a global optimum, or exact invariance under all floating-point perturbations. Uniform initialization can also remain uninformative on symmetric instances. Couplings, residuals, iteration counts, and marginal errors are retained. In the canonical study, 253 of 384 distinct target–bank comparisons terminate without reaching the 10^{-9} outer tolerance. Twenty of the 24 selected donors have such a capped comparison. Scores are reused across depths because they depend only on the two graphs. The empirical result therefore concerns this capped, regularized solver with uniform initialization; it does not establish the quality of globally optimized GW matching.

The inexpensive alternative is a six-component descriptor: mean and standard deviation of degree, its quartiles at 0.25, 0.5, and 0.75, and mean clustering coefficient. Descriptor differences are scaled by their reference-bank standard deviations, floored at 0.1, and compared in Euclidean norm. The local alternative uses d_p from Theorem 1. Neither reference choice uses the target graph’s optimized parameter label.

For scores d_j to bank graphs, let

$$w_j = \frac{\exp(-d_j/\tau)}{\sum_k \exp(-d_k/\tau)}, \quad (7)$$

$$\tau = \max\{\text{sd}(d_j), 10^{-3}\}.$$

The code subtracts the minimum score before exponentiation for numerical stability; this leaves the normalized weights unchanged. In the preserved experiments, the initial anchor x_0 is the first largest computed weight in the saved bank order. In exact arithmetic the weights decrease strictly with the scores, so this selects a score minimizer; in floating point, selection uses the computed weights without a near-tie tolerance. Rounding-scale score differences can therefore change the selected donor. For the local alternative at depth two, all 384 target–bank scores equal one to within 4×10^{-16} , so Eq. (3) is vacuous for every pair. The selected reference is nevertheless not the first bank vector on 3 of the 24 depth-two targets, and those 3 targets are the only source of the difference between the local-score and fixed-donor rows of Table 4; that difference is a rounding artifact and carries no structural information. The revised implementation selects directly from the scores, choosing the first bank index with $d_j \leq \min_k d_k + 10^{-12}$. This absolute tolerance is part of the policy and deliberately merges score differences below it; it removes the observed roundoff-dependent choice when Local TV saturates. It does not create information in a saturated score. The fixed-donor control also uses the first bank vector. Descriptor distances are the default, and the weights determine shape only after the score-based donor is selected. The archived source and records retain the historical selection rule. After symmetry alignment, only reference vectors θ_j satisfying $\|\theta_j - x_0\|_2 \leq 1$ are retained for the shape, and their weights are renormalized. Define

$$V = 0.02I + \sum_j w_j (\theta_j - x_0)(\theta_j - x_0)^\top. \quad (8)$$

Subtract the mean log eigenvalue of V , clip each resulting value to $[-\log 2, \log 2]$, and subtract their mean once more. If the resulting values are ℓ_j and the eigenvectors form U , set $B = U \text{diag}(e^{\ell_j/2}) U^\top$. Then $\det B = 1$ and $\text{cond}(BB^\top) \leq 4$. Both follow from the two recentings and the clip: the final subtraction of the mean gives $\sum_j \ell_j = 0$, so $\det B = \exp(\frac{1}{2} \sum_j \ell_j) = 1$; the clip confines the pre-centred values to an interval of length $2 \log 2$, and subtracting a constant leaves that spread unchanged, so $\max_j \ell_j - \min_j \ell_j \leq 2 \log 2$. The eigenvalues of $BB^\top$ are e^{ℓ_j} , whose ratio is therefore at most $e^{2 \log 2} = 4$. This scatter matrix is an anchor-centred scatter of the retained reference vectors, not a covariance about their weighted mean, and is used only for geometry, with no probabilistic interpretation.

4.4 Sensitivity of the transport selector

The transport sensitivity study reuses all 384 canonical target–bank pairs and their saved parameter references. It changes structural computation only: no target optimizer is rerun, and target QAOA objective values never select a solver or a start. For entropic updates, the outer limit becomes 1000 while the coupling-change tolerance remains 10^{-9} . The inner loop retains 500 scaling iterations, tolerance 10^{-12} , and the final marginal tolerance 10^{-8} . Uniform-start regularizations are 0.02, 0.05, and 0.1. A separate 0.05 multistart policy adds two seeded random positive couplings, balanced to uniform marginals, and retains the feasible returned coupling with lowest unregularized distortion. The protocol fixes seed 2026091903 and deterministic offsets by graph pair and start. The rounded cost tensor uses 12 decimal places as before. Positive-kernel scaling recenters exponents by their maximum; comparison with the original log-domain implementation is tested independently.

An additional solver applies the nonconvex conditional-gradient construction of Ref. [22] to the unregularized objective in Eq. (6). At a feasible coupling T , the nonconstant part of its gradient is $-4D_G T D_H^\top$. Each search direction solves a transportation linear program with uniform marginals using SciPy’s HiGHS interface at primal and dual feasibility tolerances of 10^{-9} ; the omitted separable gradient terms are constant along feasible directions. The step minimizes the quadratic objective along the feasible segment by comparing both endpoints and any interior minimizer. The solver uses the same three starts, at most 200 updates, and a final first-order gap tolerance of 10^{-10} , the gap being $\langle -4D_G T D_H^\top, T - S \rangle$ with S a minimizer of the same linear form over feasible couplings. The lowest-distortion returned coupling defines the comparison score. A small first-order gap is a stationarity diagnostic, not a global-optimality certificate for this nonconvex problem. On 48 of the 384 pairs the uniform start already meets the gap tolerance without any update, because two bank graphs have constant normalized distance row sums; in every such pair a random start returns strictly lower distortion. Numerically solved linear subproblems and finite precision make the reported gap a floating-point diagnostic rather than an exact stationarity certificate. The 10^{-10} gap threshold also lies below the 10^{-9} feasibility tolerance of those subproblems.

Every start’s coupling, distortion, residual or gap, iteration count, feasibility status, and marginal error is retained, including finite infeasible entropic results. The conditional-gradient routine instead raises on a returned coupling that violates the marginal tolerance, so its recorded couplings are feasible by construction rather than by observation. A target has a valid selector only when all 16 donor comparisons have feasible returned couplings; otherwise its result is recorded as unavailable. The rule does not silently discard difficult donors. Reported transfer means condition on the valid targets, whose counts are shown in Table 6; different valid-target subsets cannot support a direct ranking. For feasible targets, the lowest structural score selects a donor in fixed bank order. Its saved angles are evaluated on the target only to diagnose transfer quality. Direct four-index summation independently checks each saved distortion, and a fresh transportation subproblem checks the conditional-gradient gap. These checks support the recorded finite computations without upgrading them to exact GW optima.

4.5 Geometry controls local-model noise amplification

At a current angle vector x , write the trial region as

$$\begin{aligned} \mathcal{T}(x, \Delta, B) &= \{x + \Delta B u : \|u\|_2 \leq 1\}, \\ B &= B^\top \succ 0, \quad \det B = 1. \end{aligned} \tag{9}$$

The equivalent metric is $M = B^{-2}$. The shape B is fixed during a run; the center and radius change in response to measurements. The determinant constraint preserves volume when comparing

shapes at the same radius. Angles are evaluated modulo exact QAOA symmetries; local models are interpreted in the unwrapped coordinates of Eq. (9).

For local coordinates u_i , the affine design has rows $A_i = (1, u_i^\top)$. The model $m(u) = \hat{c}_0 + \hat{c}_{1:d}^\top u$, $d = 2p$, is fitted from shot estimates by weighted QR. The implementation always evaluates exactly $N = d + 1$ points at a common shot count, so A is square and invertible and the weighted fit reduces to interpolation of the sample means: any positive weights leave $\hat{c}$ unchanged, and no accuracy benefit is claimed for weighting. The weights are retained only because Proposition 2 is stated for $N \geq d + 1$. If $W^{1/2}A = QR$ has full column rank, then

$$\hat{c} - c^* = R^{-1}Q^\top W^{1/2}(r + e), \quad (10)$$

where c^* is the local first-order Taylor model, r is its bias at the sample points, and e is measurement error. Proposition 2 bounds each contribution under explicit smoothness and sampling conditions. The smallest singular value of R determines how strongly the fitting geometry can amplify these errors.

The implementation selects $d+1$ interpolation points by ordinary column-pivoted QR [3] applied to the transpose of a candidate design. All candidate coordinates and basis evaluations are classical; only the selected points incur measurements. This procedure preserves full model dimension. It does not perform adaptive low-rank approximation and does not claim a new QR factorization. Comparing with an unselected random design tests the value of point placement while holding the least-squares solver fixed. Ordinary pivoting alone supplies neither a strong rank-revealing guarantee nor an optimal interpolation set.

Point selection and local-model error. For dimension $d = 2p$, draw $d + 1$ independent standard Gaussian vectors and normalize each to the unit sphere. The repaired candidate set contains these vectors, 0, and the signed coordinate vectors $\pm e_j$. Column-pivoted QR of the transposed candidate design selects $d + 1$ rows. The no-repair variant uses the original random sphere points. Both check full rank and use the same QR solver. No measured objective enters point selection. The implementation records the condition number of the weighted triangular factor, not the smallest singular value $\sigma_{\min}(R)$ that Eq. (13) inverts; a condition number alone does not bound the inverse-design sensitivity. For the repaired design the pivoted selection returned the same point set in every recorded trial, so this quantity is in fact constant: at the canonical 256 shots per point, $\sigma_{\min}(R) = 12.25$ at $p = 1$ and 9.52 at $p = 2$, while the recorded condition number is correspondingly fixed at $1 + \sqrt{2} \approx 2.414$ and 3.992. With $\delta = 0.05$ the stochastic term of $\mathcal{E}$ is then about 0.22 and 0.38, which is not small against the unit range of the normalized objective. Interpolation geometry and inverse-design bounds have an established role in local-model accuracy [5, Sections 3–5]. The following proposition combines a Taylor remainder, weighted least-squares perturbation, and bounded-observation concentration for the specified noisy affine model; full rank alone does not impose all conditions of a fully linear model.

Proposition 2 (Local model error with bounded observations). *Suppose ∇f is L_f -Lipschitz on a convex set containing $\mathcal{T}(x, \Delta, B)$. Choose $N \geq d + 1$ points with $\|u_i\|_2 \leq 1$ and positive shot counts S_i before drawing the stage's independent observations in $[-1, 0]$, each with conditional mean $f(x + \Delta B u_i)$ at point i . Let $W = \text{diag}(S_i)$ and let $W^{1/2}A = QR$ have full column rank. Define*

$$c^* = (f(x), \Delta B^\top \nabla f(x))^\top, \quad (11)$$

$$a_i = \frac{1}{2} L_f \Delta^2 \|B u_i\|_2^2, \quad \sigma = \sigma_{\min}(R) > 0. \quad (12)$$

For any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\|\hat{c} - c^*\|_2 \leq \frac{\sqrt{\sum_i S_i a_i^2} + \sqrt{\frac{N}{2} \log(2N/\delta)}}{\sigma} =: \mathcal{E}. \quad (13)$$

On the same event, uniformly for $\|u\|_2 \leq 1$,

$$|f(x + \Delta B u) - m(u)| \leq \frac{1}{2} L_f \Delta^2 \|B\|_2^2 + \sqrt{2} \mathcal{E}. \quad (14)$$

These statements also hold conditionally when the design and shot counts are chosen from the history preceding this stage.

Proof. The first-order Taylor remainder satisfies $|r_i| \leq a_i$. Writing the vector of sample means as $y = A c^* + r + e$ gives Eq. (10), since $\hat{c} = R^{-1} Q^\top W^{1/2} y$ and $Q^\top Q = I$. Hoeffding's inequality [18] and a union bound give

$$|e_i| \leq \sqrt{\frac{\log(2N/\delta)}{2S_i}} \quad \text{for every } i \quad (15)$$

with probability at least $1 - \delta$. Therefore $\|W^{1/2} r\|_2 \leq \sqrt{\sum_i S_i a_i^2}$ and $\|W^{1/2} e\|_2 \leq \sqrt{N \log(2N/\delta)/2}$. Using $\|R^{-1} Q^\top\|_2 = 1/\sigma$ proves Eq. (13). At any point in the region, the Taylor remainder is at most $L_f \Delta^2 \|B\|_2^2/2$ and $\|(1, u^\top)\|_2 \leq \sqrt{2}$. The triangle inequality proves Eq. (14). Conditioning fixes all choices made before the independent stage observations, so the same argument applies to a history-dependent design. $\square$

For QAOA the objective is analytic, so the required smoothness holds on any compact region, but the theorem does not supply a small useful L_f . Equal shot counts S make the stochastic term proportional to $S^{-1/2}/\sigma_{\min}(A)$; the bias term need not decrease with S . Because the gradient block of c^* is $\Delta B^\top \nabla f(x)$, the recovered gradient is $\Delta^{-1} B^{-\top} \hat{c}_{1:d}$, whose stochastic error scales as $\Delta^{-1} S^{-1/2}$. Contracting the radius from the canonical 0.4 to the floor 0.025 is a factor of 16, so 256 times the shots per point would preserve this stochastic gradient-error bound at the same design and shape. In the recorded runs 26.2% of radius-adapting trials are taken at the floor radius, and none of the 240 accepted steps occurs at a radius below 0.15. The historical model uses $S = 256$ at each selected point at every radius and does not estimate L_f or choose S by enforcing Eq. (14). Consequently, the proposition explains sources of model error without certifying every fitted model as sufficiently accurate for convergence.

The revised policy instead chooses, before any model observations,

$$S(\Delta) = \max\{2, \lceil S_0(\Delta_0/\Delta)^2 \rceil\}. \quad (16)$$

For fixed design and shape this keeps the stochastic part of recovered gradient error at a fixed scale, up to integer rounding, as the region contracts. It does not make the error proportional to Δ . With bounded design sensitivity, the stronger stochastic accuracy needed for a fully linear model generally is obtained from this bound with S proportional to Δ^{-4} , apart from confidence factors: coefficient noise must be $O(\Delta^2)$ so that physical gradient noise is $O(\Delta)$. Neither schedule removes affine Taylor bias. Figure 4 therefore measures both the sampling error about the exact-data interpolant and the total error relative to the true objective gradient.

Predicting bias and propagated measurement variance. Condition on the history before fitting, so that the design, shape, radius, weights, and shot counts are fixed. Write $\bar{y}_i = f(x + \Delta B u_i)$, $y_i = \bar{y}_i + e_i$, and $v_i = \text{Var}(Y_i)$ for an individual measurement at design point i . Independent pointwise streams give $\mathbb{E}e = 0$ and $\Sigma_y = \text{diag}(v_i/S_i)$. With $P = (0, I_d)$ selecting the gradient coefficients, define the physical-gradient fitting map

$$L_A = \Delta^{-1} B^{-\top} P R^{-1} Q^\top W^{1/2}, \quad \hat{g} = L_A y, \quad \bar{g} = L_A \bar{y}. \quad (17)$$

For the square interpolation designs used here, the coefficient map is A^{-1} ; the implementation obtains it by a stable linear solve, without forming normal equations. The exact gradient $g = \nabla f(x)$ need not equal $\bar{g}$. Their difference is the finite-radius affine-model bias.

Proposition 3 (Conditional gradient mean-square error). *Under the stated conditional unbiasedness and finite-variance assumptions,*

$$\mathbb{E}[\|\hat{g} - g\|_2^2 \mid \mathcal{H}] = \|\bar{g} - g\|_2^2 + \text{tr}(L_A \Sigma_y L_A^\top). \quad (18)$$

The corresponding sampling-error mean square is $\mathbb{E}[\|\hat{g} - \bar{g}\|_2^2 \mid \mathcal{H}] = \text{tr}(L_A \Sigma_y L_A^\top)$.

Proof. Write $\hat{g} - g = (\bar{g} - g) + L_A e$. Expanding the squared Euclidean norm leaves the deterministic bias term and $\mathbb{E}[e^\top L_A^\top L_A e \mid \mathcal{H}]$. The cross term vanishes because $\mathbb{E}[e \mid \mathcal{H}] = 0$. The remaining expectation is $\text{tr}(L_A \mathbb{E}[e e^\top \mid \mathcal{H}] L_A^\top)$, which is the stated variance term. The sampling-error identity follows by replacing g with $\bar{g}$. $\square$

This is the standard bias–variance decomposition for a linear estimator, not a new concentration inequality or QAOA theorem. Its testable role is to predict the magnitude and radius dependence of errors before new model shots are drawn. We record $\|L_A\|_2$ and $\|L_A\|_F$ alongside the weighted-design condition number: the latter alone cannot quantify physical gradient sensitivity. RMS curves take the square root after averaging squared errors, rather than averaging norms.

With constant design and shape and equal shots, the variance term is

$$\Delta^{-2} S^{-1} \text{tr}(B^{-\top} P A^{-1} \text{diag}(v_i) A^{-\top} P^\top B^{-1}).$$

Equation (16) holds its explicit inverse-radius factor fixed, but the pointwise variances can still change with radius. It does not control the bias term or the nonlinear step direction. For $(S_0, \Delta_0) = (256, 0.1)$, the actual schedule at radii 0.025, 0.05, 0.1, 0.2 and 0.4 is 4096, 1024, 256, 64 and 16 shots per point; all five operating points are measured in the new diagnostic.

The prediction uses exact simulator means, variances, and a numerically validated reference gradient. These quantities are fixed before the independent observations used to test it, including on fresh targets, but are not assumed available to a deployed finite-shot optimizer. A Gaussian approximation to the affine coefficients is tested separately for positive proposal margins because Eq. (18) alone does not predict the nonlinear normalized boundary step or the power of its endpoint test.

Adaptive radii do not invalidate Proposition 2. Conditional on the preceding history, Eq. (16) and the design are fixed before the independent model samples. For a simultaneous model statement over at most T stages, apply the proposition with $\delta_t = \delta/T$ at each possible stage and union bound after taking conditional expectations. Model and endpoint streams are separate. Intersecting this model event with the acceptance event gives probability at least $1 - \delta - \alpha$ without requiring independence between the two events. The implementation does not estimate the smoothness constant or claim that this event certifies the accuracy of every realized model.

Equation (14) also links model quality to the acceptance rule of the next subsection. Write $\mathcal{E}_f = \frac{1}{2}L_f\Delta^2\|B\|_2^2 + \sqrt{2}\mathcal{E}$ for the uniform model error, let $q = \|\hat{c}_{1:d}\|_2$ be the decrease predicted by the model's boundary step, and let a be the true decrease at that step. Applying Eq. (14) at the center and at the proposed point gives $a \geq q - 2\mathcal{E}_f$, so the detection condition $a - \eta q \geq 4b_L$ of Corollary 5, with acceptance slack $\eta \in (0, 1)$ and confidence allowance b_L from Eq. (20), is implied by

$$(1 - \eta)q \geq 2\mathcal{E}_f + 4b_L. \quad (19)$$

Provided the last prescribed look is affordable, this is a sufficient joint condition on proposal quality and measurement resolution, on the intersection of the model and acceptance confidence events. Failure of the condition does not rule out a useful or accepted step: the error bounds may be conservative. Neither $\mathcal{E}_f$ nor L_f is estimated in the runs, so Eq. (19) provides an analytical connection rather than an implemented test.

4.6 Accepted steps have a finite-horizon statistical guarantee

The affine model proposes the boundary step $u^+ = -\hat{c}_{1:d}/\|\hat{c}_{1:d}\|_2$ when its gradient is nonzero. Its predicted decrease is $q = m(0) - m(u^+) = \|\hat{c}_{1:d}\|_2 > 0$. Fresh measurement streams estimate the actual decrease $a = f_G(x) - f_G(x + \Delta Bu^+)$. At prespecified cumulative sample sizes $s_1 < \dots < s_L$, define

$$b_\ell = \sqrt{\frac{\log(4TL/\alpha)}{2s_\ell}}, \quad \hat{a}_\ell = \bar{Y}_{x,\ell} - \bar{Y}_{+, \ell}, \quad (20)$$

where T bounds the number of trials and α is the failure probability for one optimization run. A trial is accepted only if

$$\hat{a}_\ell - 2b_\ell \geq \eta q, \quad 0 < \eta < 1. \quad (21)$$

Theorem 4 shows that, with probability at least $1 - \alpha$, every accepted step in that run satisfies $a \geq \eta q > 0$, despite adaptive trial points and early stopping among the prescribed looks. Model accuracy is not needed for this acceptance statement, because acceptance uses fresh bounded observations.

This separation is useful but has a cost. A poor model can propose many unacceptable steps, and a small true improvement can remain unresolved within the allowed shots. A trial whose interval still intersects the threshold at the last affordable look is recorded as unresolved and the incumbent is retained. Such a decision is not evidence that the trial is worse. With $T = 64$, $L = 4$, $\alpha = 0.05$, and the final look $s_L = 32768$, the two-endpoint confidence allowance $2b_L$ is approximately 0.0246 in normalized objective units. Reaching that look costs 65536 shots for the endpoint pair, so it is unaffordable at the 32768-shot budget; in the canonical study it is reached by 1,440 of the 21,196 trials that follow the four prescribed looks, all at the larger budget, while the deepest look reached at the smaller budget is 8192 shots per endpoint, where the allowance $2b_\ell$ is approximately 0.0492. This statistical scale must be considered when interpreting small expected improvements.

Simultaneous validity and detectable improvements.

Theorem 4 (Simultaneous validity over prescribed looks). *Consider at most T adaptive trials. Before a trial's acceptance measurements, its center, trial point, and predicted decrease $q > 0$ are fixed conditional on the previous history. At each endpoint, start a fresh stream of conditionally independent observations in $[-1, 0]$ with the corresponding true objective mean. Use only the L cumulative sample sizes specified in advance and the bounds in Eq. (20). With probability at least*

$1 - \alpha$, all endpoint confidence intervals at all executed trials and looks are valid. On this event every acceptance by Eq. (21) satisfies

$$f_G(x) - f_G(x + \Delta B u^+) \geq \eta q > 0. \quad (22)$$

Early stopping or budget-based omission of a prescribed look does not change the guarantee.

Proof. Condition on the history before one trial. For a fixed endpoint and look, Hoeffding's inequality gives error probability at most $2 \exp(-2s_\ell b_\ell^2) = \alpha/(2TL)$. A union bound over two endpoints and L looks bounds that trial's conditional failure probability by α/T . Taking expectations and union bounding over at most T trials gives a total failure probability at most α . The argument uses fresh streams after trial selection; within a trial, cumulative estimates may share observations. On simultaneous validity, $|\hat{a}_\ell - a| \leq 2b_\ell$, so Eq. (21) implies $a \geq \eta q$. Stopping selects a subset of the controlled events and therefore cannot increase this probability bound. $\square$

Corollary 5 (A sufficient measurement scale for detection). *Under the assumptions of Theorem 4, fix a trial with true decrease a and predicted decrease q . If its last prescribed look is affordable and*

$$a - \eta q \geq 4b_L > 0, \quad (23)$$

then on the simultaneous confidence event it is accepted at or before that look. Equivalently, for a margin $h = a - \eta q > 0$, the sufficient final sample count per endpoint is

$$s_L \geq \frac{8 \log(4TL/\alpha)}{h^2}. \quad (24)$$

Proof. On the stated event $|\hat{a}_\ell - a| \leq 2b_\ell$ at every look. No earlier rejection is possible when $a > \eta q$, since $\hat{a}_\ell + 2b_\ell \geq a > \eta q$. At the final look, $\hat{a}_L - 2b_L \geq a - 4b_L \geq \eta q$, which gives acceptance. Substituting Eq. (20) and squaring positive quantities proves Eq. (24). $\square$

The sufficient margin is about 0.0492 for the canonical final look. Even if every observed mean equalled its expectation, the canonical rule would still require $a - \eta q \geq 2b_L \simeq 0.0246$ at that look. This last statement describes the decision threshold in the absence of sampling error, not an impossibility bound for random observations. The corollary is a detection guarantee conditional on a sufficiently good proposal; it does not ensure that the local model produces one. Its inverse-square dependence on the improvement margin helps explain why a statistically valid rule can spend a large budget without certifying useful refinement near a good anchor: by Eq. (24), the sufficient detection guarantee at $h = 0.005$ uses about 3.18×10^6 shots per endpoint, roughly twenty-four times the largest per-run optimization budget in this study, and about forty-eight times it for the endpoint pair.

Proposition 6 (Acceptance probability below the confidence allowance). *Fix the history $\mathcal{H}$ after selecting a trial and before drawing its acceptance samples. Let $h = a - \eta q$ be its true acceptance margin. Suppose all observations within and across the two fresh endpoint streams are conditionally independent given $\mathcal{H}$, lie in $[-1, 0]$, and have the corresponding true objective means. For the Hoeffding rule in Eq. (21),*

$$\mathbb{P}(\text{accept this trial} \mid \mathcal{H}) \leq \min \left\{ 1, \sum_{\ell=1}^L \exp \left[-s_\ell (2b_\ell - h)_+^2 \right] \right\}, \quad (25)$$

where $(t)_+ = \max\{t, 0\}$. The bound remains valid when early rejection or an insufficient budget prevents some prescribed looks.

Proof. At look ℓ , $\hat{a}_\ell - a$ is a centered sum of $2s_\ell$ independent terms, each with range length $1/s_\ell$. The sum of squared range lengths is $2/s_\ell$. The one-sided form of Hoeffding’s inequality [18] therefore gives $\mathbb{P}(\hat{a}_\ell - a \geq t \mid \mathcal{H}) \leq \exp(-s_\ell t^2)$ for $t \geq 0$. Acceptance requires $\hat{a}_\ell - a \geq 2b_\ell - h$. Apply this inequality when the threshold is positive and use the trivial bound one otherwise. Union bound over the prescribed looks, extending the fresh streams to unexecuted looks if necessary. Removing looks cannot add acceptance events, and independence between cumulative looks is unnecessary. $\square$

For the canonical four-look schedule, every trial with $h \leq 0.005$ has conditional acceptance probability at most 3.48×10^{-6} , even if all four looks are affordable. This is an upper bound for the specified policy; it does not constrain other estimators or acceptance rules. Together with Corollary 5, it distinguishes valid decisions from a rule’s ability to detect small improvements.

The canonical policy allows $T = 64$ trials with $\eta = 0.1$ and $\alpha = 0.05$. It examines each endpoint after 512, 2048, 8192, and 32768 cumulative shots. New batches contribute only the difference from the preceding look. If $\hat{a}_\ell + 2b_\ell < \eta q$, the trial is rejected for failing the required decrease on the confidence event. Otherwise additional shots are requested until a decision or a budget limit. Unresolved trials leave the center unchanged. Confidence is per optimization run; 95% does not mean simultaneous validity across the complete collection of benchmark runs. Methods and budgets reuse seed labels to support paired comparisons; their outcomes are not assumed independent.

The canonical initial radius is 0.4. Acceptance multiplies it by 1.5 up to 0.8; rejection or an unresolved decision multiplies it by 0.5 down to 0.025. A predicted decrease below 10^{-12} terminates the run. A fixed-radius ablation retains 0.4 throughout. The fixed-shots ablation uses only the final 32768-shot look, with $L = 1$ in the confidence bound. This is a prespecified sampling policy, not a hindsight-matched optimal fixed allocation. Its single look costs 65536 shots for the endpoint pair, so no run affords a second trial at either canonical budget: at 32768 shots all 144 fixed-shots runs fit one model and stop before any acceptance measurement, and at 131072 shots all 144 complete exactly one acceptance test, every one of which rejects. This ablation therefore accepts no step anywhere in the study. Such costs and unresolved decisions are retained in the records.

Algorithm: resolution-aware refinement after transfer. The revised routine `refine` takes an anchor x_0 , a fixed shape B , a shot cap $N_{\max}$, initial radius Δ_0 , initial model count S_0 , and prescribed acceptance looks. In the revision experiment, $(S_0, \Delta_0) = (256, 0.1)$, $T = 16$, $\eta = 0.1$, $\alpha = 0.05$, and $s = (512, 2048, 8192, 16384)$, with radius bounds $[0.025, 0.8]$. The input shape is I or the bounded descriptor shape. Its output is the last accepted incumbent, an explicit stopping reason, and the full ledger.

1. Initialize $x = x_0$, $\Delta = \Delta_0$, and spent shots to zero. Record the anchor as the zero-shot incumbent. Seed separate streams for classical point selection, model sampling, and endpoint sampling.
2. Before trial t , compute $S(\Delta)$ by Eq. (16). If the remaining shots cannot cover $(d+1)S(\Delta) + 2s_1$, stop with `resolution_limited` and reason `model_and_first_look_unaffordable`. Spend no model shots for that untestable proposal. Reserving the first look assures feasibility of one test; it does not assure detection at that look.
3. Select a full-rank QR design, sample its $d + 1$ points at $S(\Delta)$, and fit the affine model. Record radius, design, coefficients, shot count, and condition number. If $q = \|\hat{c}_{1:d}\|_2 < 10^{-12}$, return the incumbent with a flat-model resolution reason. Otherwise propose $x^+ = \text{wrap}(x - \Delta B \hat{c}_{1:d}/q)$.

4. At each prescribed look request fresh observations at both endpoints, adding only $s_\ell - s_{\ell-1}$ samples per endpoint. Reject an unaffordable batch atomically. Compute the decrease interval using either Eq. (20) or Eq. (26). Accept if its lower endpoint is at least ηq , and reject if its upper endpoint is below ηq . Otherwise continue to the next affordable look.
5. After acceptance set $x = x^+$, record its cumulative shot count, and expand Δ by 1.5 up to 0.8. Only a certified rejection contracts Δ by 0.5. Rejection at the radius floor terminates with `radius_limit`. An unresolved decision stops immediately with `resolution_limited`, recording whether the prescribed looks or remaining budget were exhausted; it leaves both the incumbent and radius unchanged.
6. Continue to the declared horizon unless a stopping condition applies. Return the incumbent at every ordinary stop. Rank or nonfinite-input failures raise explicit errors rather than silently substituting a model.

This policy distinguishes insufficient measurement evidence from evidence against sufficient decrease. A resolution stop is operational: it says that the declared model or decision schedule has ended without resolving a further update under this policy. It is not a proof that all other proposals, estimators, or allocations would fail. On the simultaneous confidence event, a certified rejection proves $h < 0$, which still permits a positive improvement $a > 0$. The preserved `optimize` and `bernstein_optimize` routines retain the historical contraction after unresolved decisions. They are used only for their explicitly labelled controls and replay.

For attribution, `model_sampling=fixed` replaces Eq. (16) by $S = S_0$. The orthogonal control `unresolved_policy=continue` retains the radius after an unresolved test and fits a new model if the same preflight reservation is affordable. The default remains radius-dependent sampling and immediate unresolved stopping. The continued policy differs from the historical routine, which also contracted the radius after an unresolved test. All four attribution cells use separate but identically seeded design, model, and endpoint streams, giving identical first-trial observations before any policy-driven divergence. Fixed and adaptive schedules are compared at the returned incumbent, never by retrospectively selecting a better proposal.

The stop status and its reason are reported separately. A failed preflight means the remaining total budget cannot fund the prescribed model and first look. An unaffordable later look is a different total-budget limit. An unresolved test at its last prescribed look is a policy cap, even when additional total-budget shots remain. A certified rejection is evidence against $a \geq \eta q$ on the confidence event, while flat-model termination is a numerical/model threshold. None of these statuses alone establishes stationarity or the impossibility of a useful alternative allocation.

4.7 Canonical instances, reference bank, and comparators

The canonical configuration fixes the master seed to 20260918. NetworkX generators [16] produce independent-edge $G(n, 0.4)$ graphs [13], random 3-regular graphs using the Steger–Wormald generator [33], Barabási–Albert attachment graphs with two new edges per vertex [1], and Watts–Strogatz rewiring graphs with initial degree four and rewiring probability 0.35 [37]. Disconnected draws are rejected. The reference bank contains two graphs from each family at each of $n = 8, 10$, giving 16 graphs. The test set contains three per family at each of $n = 10, 12$, giving 24 graphs. Every accepted graph is nonisomorphic to all earlier accepted graphs in either split, including across family labels. These filters modify the base sampling distributions; exact uniformity over the retained regular graphs is not assumed. Saved edge lists are the definitive instances. The study examines $p = 1, 2$ separately, with a separate parameter bank at each depth.

Table 3: Prespecified comparators and ablations. All local-model variants share the QR solver and finite-horizon acceptance rule except where a sampling or radius change is stated.

Variant	Controlled choice
Fixed donor	First bank anchor, isotropic shape
Random start	Uniform random angle vector in the symmetry cell, isotropic shape
Descriptor, isotropic	Descriptor-selected anchor, $B = I$
Descriptor, shaped	Same anchor, reference scatter shape
GW, isotropic	GW-selected anchor, $B = I$
GW, shaped	Same anchor, reference scatter shape
Local TV	Rooted-neighborhood total-variation score, $B = I$
Shuffled GW	Permuted GW scores before anchor and shape
Displaced prior	Descriptor anchor shifted by $(1.2\mathbf{1}_p, 0.35\mathbf{1}_p)$ and wrapped, reference scatter shape
No point repair	Descriptor shape, random interpolation points
Fixed radius	Descriptor shape, radius fixed to 0.4
Fixed shots	Descriptor shape, one final acceptance look
COBYLA	Descriptor anchor, 1024 shots per estimate
SPSA	Descriptor anchor, 1024 shots per estimate

For each graph and depth, six BFGS runs [26] on the exact expectation supply a numerical reference: one deterministic angle schedule and five uniformly randomized starts in the symmetry cell. The SciPy implementation [36] estimates gradients by two-point finite differences because no Jacobian is supplied. BFGS uses at most 180 iterations and gradient tolerance 10^{-7} . For every start we retain the returned angle vector, its objective value, the convergence flag, the termination message, and the exact evaluation count, together with one construction time per graph and depth. The starting vectors themselves are not stored; they are fixed by the recorded configuration and seed. The lowest recorded value is f_G^{ref} ; it is not asserted to be globally optimal. Test-graph references are diagnostic only and are not available to the optimizers or graph-prior construction. The signed reference gap $f_G(\theta) - f_G^{\text{ref}}$ may be negative.

Fourteen methods are specified before measurement trials. Table 3 gives their controls. All isotropic/shape pairs use the same selected anchor. The shape, no-repair, fixed-radius, and fixed-shots variants share the descriptor-selected anchor unless explicitly identified as GW variants. For each graph, depth, and method, three shot seeds are run at optimization budgets 32768 and 131072. All accepted and unsuccessful trials count toward the budget. The public configuration determines the complete study rather than a selection of successful runs.

COBYLA uses Powell’s derivative-free method [29] through the PRIMA implementation [39] included in SciPy 1.17.1 [36], with initial radius 0.4 and tolerance 0.002, subject to the same shot cap. Its `maxiter` option bounds objective evaluations. SPSA uses the simultaneous-perturbation gradient estimator [32] with independent Rademacher perturbations. We set the gains to $a_k = 0.15/(k+10)^{0.602}$ and $c_k = 0.12/k^{0.101}$ and use two perturbed estimates plus one additional incumbent estimate per update. Both retain the point with the smallest observed incumbent estimate. Their true-objective trajectories need not be monotone. These baseline settings are fixed rather than tuned on the test results; a different competent tuning may change their ranking.

4.8 Follow-up across two reference banks

The follow-up is a separate experiment designed after examining the canonical study. Its full configuration is `code/configs/followup.json`. Two panels use seeds 2026091901 and 2026091902. Each contains eight bank graphs (the four families above at $n = 8, 10$), four validation graphs (one per family at $n = 12$), and eight test graphs (the four families at $n = 12, 14$). All 40 accepted graphs are mutually nonisomorphic. Cross-study inspection finds one 12-vertex regular follow-up test graph isomorphic to a canonical test graph; the other 15 test graphs are nonisomorphic to every canonical instance. This overlap is retained and disclosed rather than removed after observing outcomes. One eight-vertex regular bank graph also matches a canonical bank graph; the other 15 bank graphs and all eight validation graphs are nonisomorphic to every canonical instance. No follow-up bank or validation graph matches a canonical test graph. Every target uses only its own panel’s bank. The two panels are separately seeded replications with an isomorphism exclusion across panels, rather than unrestricted independent graph draws.

The circuit depths are $p = 2, 3$. References use six BFGS starts, tolerance 10^{-7} , and at most 120 iterations per start. As in the canonical study, these are best-found references, with every exact objective call retained; no global-optimality claim is made. The seven test policies are descriptor initialization alone, a fixed graph-independent schedule, matched isotropic Hoeffding refinement, isotropic and shaped empirical-Bernstein refinement, SPSA, and COBYLA. The fixed schedule is $\gamma = \text{linspace}(0.35, 0.8, p)$ and $\beta = \text{linspace}(0.35, 0.15, p)$. Initialization-only and fixed-schedule policies make no optimization calls. All other policies start from the same descriptor-selected anchor. Only the shaped Bernstein variant uses $B \neq I$.

Each run has an optimization cap of 65536 shots and receives 8192 independent final-validation shots. At each depth, the validation phase compares four combinations of model shots $S \in \{256, 1024\}$ and initial radius $\Delta \in \{0.1, 0.3\}$, using both Bernstein shapes on all eight validation graphs. This gives 128 validation runs. The selection criterion is the mean finite-shot final-validation objective over graphs and the two shapes; exact expectations and test outcomes are not used. Candidate order breaks ties. All validation runs exhaust the optimization budget without accepting a trial, so each returns its initial anchor; because the final-validation seed is derived from the run seed, which does not depend on the candidate, the four candidates draw the same 8192 shots at the same angle and the four scores are bitwise identical at each depth. Thus $S = 256$ and $\Delta = 0.1$ are selected by the tie-break at both depths, not by an observed quality improvement. The selected settings and validation records are hashed before the 448 test runs: seven methods, 16 graphs, two depths, and two shot seeds. Paired methods reuse seed labels; their results are not independent within a graph. The shared local-model settings enable a controlled comparison of acceptance rules at $B = I$. No shaped Hoeffding policy was run in this historical follow-up, so its two-by-two design of isotropic and shaped regions crossed with Hoeffding and empirical-Bernstein acceptance is missing one cell. The acceptance rule is therefore varied only at $B = I$ and the shape only under empirical-Bernstein acceptance; the shape-by-rule interaction is not identified, and that experiment leaves the shaped Hoeffding cell untested. The revision below completes all four cells under the revised stopping and allocation policy. SPSA retains its gains, and both SPSA and COBYLA use 1024 shots per estimate. COBYLA retains tolerance 0.002 but uses the common selected initial radius 0.1, instead of the canonical value 0.4. Neither baseline receives a separate tuning search.

The follow-up retains $\eta = 0.1$ and $\alpha = 0.05$, uses $T = 16$ trials, and prescribes cumulative looks (512, 2048, 8192, 16384) per endpoint. The radius contracts by 0.5 or expands by 1.5 with the same bounds $[0.025, 0.8]$. Fresh endpoint streams are sampled after the model proposes a trial. For an endpoint with $s \geq 2$ observations, let $\hat{v}_s$ be the unbiased sample variance of individual

outcomes. Assume the observations at each fixed endpoint are conditionally i.i.d.; this holds for the ideal multinomial measurement model used here. The established empirical Bernstein inequality of Maurer and Pontil [23, Theorem 4] gives the radius

$$r_s = \sqrt{\frac{2\hat{v}_s \log(8TL/\alpha)}{s}} + \frac{7 \log(8TL/\alpha)}{3(s-1)}. \quad (26)$$

Apply their one-sided inequality to $Y + 1 \in [0, 1]$ and its reflection, assigning failure probability $\alpha/(4TL)$ to each endpoint and tail. Conditioning on the history before each trial and taking a union bound over all prescribed trials and looks gives simultaneous interval validity with probability at least $1 - \alpha$ per run. The decrease interval has radius $r_{x,s} + r_{+,s}$: acceptance requires $\hat{a}_s - r_{x,s} - r_{+,s} \geq \eta q$, and rejection requires $\hat{a}_s + r_{x,s} + r_{+,s} < \eta q$. Other outcomes remain unresolved. This is a direct finite-horizon application of an existing inequality, not a new confidence bound. It preserves the true-decrease conclusion of Theorem 4 with the stated conditional i.i.d. assumption at each endpoint.

Cumulative sample variances combine each batch’s within-batch sum of squares with the between-batch correction. Averaging the recorded variances of batch means would not give the required $\hat{v}_s$. The comparator models uncertainty in acceptance measurements; it does not fit a spatial variance model or reproduce ASTRO-DF [15]. The targets $f_G^{\text{ref}} + \epsilon$ for $\epsilon \in \{0.002, 0.005, 0.01, 0.02\}$ are fixed in the follow-up configuration before its test phase. Their hitting costs use recorded incumbents and exact diagnostic expectations; they never stop or steer the optimizer. Table 5 summarizes the complete test sample. The two-bank structure and small graph sample restrict generalization; per-panel summaries accompany pooled graph-level intervals. Each panel draws its bank, validation split, test split and random streams together, so a difference between panels is not attributable to the reference bank alone.

4.9 Revision experiment and frozen-proposal diagnostics

The revision fixes its protocol before drawing its new samples and records it in `code/results/refinement.json.gz`. It reuses the historical follow-up targets and their saved descriptor anchors, shapes, and best-found references. All four revised policies use two shot repetitions on all 16 targets at each of $p = 2, 3$, giving 256 new optimizer runs at a cap of 65536 shots. Settings come from the previously tied validation search; no test outcome selects new hyperparameters. Historical initialization-only, SPSA, COBYLA, and local-model records supply matched controls only where graph, depth, anchor, and budget agree. Their outcomes remain labelled historical. Reusing these instances after inspecting the earlier results makes the revision exploratory, even though its new sampling streams are fixed in advance.

The first revision’s model diagnostic uses the first panel’s eight test graphs, both depths, both shapes, radii 0.025, 0.05, 0.1, 0.2, 0.4, and model counts 256, 1024, 4096, 16384, with four independent sample seeds. For each fixed design it fits both exact objective values and independent sample means. The physical fitted gradient is $\Delta^{-1} B^{-\top} \hat{c}_{1:d}$. Central differences of the exact objective with step 10^{-5} supply a retrospective gradient reference; this is a numerical diagnostic rather than a symbolic derivative. Sampling error compares sampled and exact-data fits, whereas total error compares the sampled fit with that derivative. True proposal decrease and margin are evaluated after fitting. The exact-data control can expose a poor affine direction or excessive boundary step even when sampling noise is absent; more endpoint shots cannot repair that proposal.

For the first revision’s power experiment, the protocol freezes all 160 proposals at $S = 1024$ and sample replicate zero: eight graphs, two depths, two shapes, and five radii. Each receives 128 independent endpoint replays under both rules at caps 16384 and 65536 per endpoint. The

common planned looks are (512, 2048, 8192, 16384, 65536) with $T = 16$, $L = 5$, and $\alpha = 0.05$; the smaller cap omits the final look without changing the error allocation. The largest pair costs 131072 shots. These are offline decision-power experiments, distinct from the optimizer’s four-look, 65536-shot-cap runs. Both confidence rules use the same horizon, looks, cap, and error allocation. Independent replays report acceptance, rejection, unresolved frequency, and actual acceptance-stage expenditure, charging unresolved cases through their caps. No proposal is selected according to its observed replay success. Binning or ordering by retrospective h is for presentation only. Finite Monte Carlo frequency zero does not imply probability zero, and the analytic Hoeffding upper bound concerns the declared rule rather than all possible QAOA measurements.

Budget-indexed outputs are reconstructed from completed incumbent records. Between recorded updates the previous incumbent is retained, including during an unfinished measurement batch. Target fractions include the zero-shot anchor and retain unsuccessful runs. A first-hit fraction and a fraction whose current output meets the target answer different questions for non-monotone baseline trajectories; figure labels specify the quantity. Independent output-evaluation costs are reported separately. The diagnostic model grid uses 83,558,400 simulated observations and the endpoint replay generates 2,684,354,560 more, reusing each frozen stream across rules and caps. These offline diagnostic samples are excluded from optimizer curves and are not hardware executions. Exact references, derivative diagnostics, and replay computations are classical analysis expenses, not free oracle information available to the optimizer.

4.10 Attribution and predictive-validation protocol

The mechanism study is saved separately in `code/results/mechanism.json.gz`. It links to the unchanged historical and first-revision bundles and embeds its own protocol and numerical source. The attribution grid uses the 16 existing test graphs, depths two and three, two shot repetitions, and the four sampling-by-stopping cells described above, for 256 additional optimizer runs. All use descriptor anchors, descriptor-shaped regions, Bernstein acceptance, and a 65536-shot cap. The common settings remain $S_0 = 256$ and $\Delta_0 = 0.1$; neither the revised test outcomes nor the new observations retune them. This factorial isolates the two declared interventions within this fixed implementation, not every possible continuation or adaptive-sampling policy.

The new conditional-error grid uses the original eight diagnostic targets and eight fresh targets, the four families at sizes 12 and 14. Every fresh graph is checked for isomorphism against all previously used graphs and against the other fresh graphs. Fresh targets use the fixed first-panel donor bank and deterministic descriptor selection. The fresh split tests a prediction fixed after examining earlier targets; it is not an independent optimizer benchmark or a structural-shift experiment. All graph identities, seeds, generation filters, and selected donors are saved.

The grid crosses both depths, both shapes, the same five radii, and shot counts 16, 64, 256, 1024, 4096 and 16384. Each cell uses 128 independent sampled fits. Before those observations, exact design-point means and variances specify Eq. (18). Central differences at steps 10^{-5} and 5×10^{-6} are compared at every anchor; both vectors and their discrepancy are recorded. This validation checks derivative resolution within the exact simulator, not hardware gradient measurement. The fixed linear map, bias, propagated variance, and sensitivity norms are retained.

Along the actual radius-dependent schedule, a Gaussian coefficient model with the exact fitted mean and propagated coefficient covariance supplies 256 prespecified proposal draws on each fresh-graph cell. Their true QAOA margins give an approximate positive-margin prediction, which is frozen before 32 independent multinomial model fits are evaluated at that operating point. The Gaussian approximation need not be accurate at 16 or 64 shots and its calibration is measured, not assumed. Both prediction and observation use exact retrospective proposal objectives; their

agreement would support this conditional mechanism, not identify a measurement-only optimizer.

The four-look power analysis reuses the stored independent endpoint observations at cumulative sizes (512, 2048, 8192, 16384), recomputing every radius with $L = 4$, $T = 16$, and $\alpha = 0.05$. Merely truncating a five-look calculation would leave the wrong confidence penalty. Its budget axis is the endpoint-pair allowance; the actual online run must additionally pay its model cost. The five-look extension uses its own $L = 5$ penalty and is reported separately. Positive-margin and nonpositive-margin cohorts are fixed before varying budget. The same proposals and weights enter every point of each curve; no budget-dependent selection creates apparent power.

4.11 Resource accounting, validation, and reproducibility

An objective-estimation call is one newly sampled circuit estimate at one angle vector. A shot is one repeated circuit execution. A job is one logical batch submitted by the algorithm; a model batch contains all interpolation points, and an acceptance batch contains the two endpoints. Additional samples at an old point constitute a new estimation call. The oracle rejects a batch atomically if it would exceed the remaining shot budget. It records every call, batch, shot count, angle, sample mean, estimated variance of the mean, and elapsed simulation time. Budget caps are limits, not an assertion that every run spends exactly the same number of shots.

In the canonical study, every final vector receives 16384 independent validation shots, recorded as one additional call and job outside the optimization budget. Total operational shot cost is the optimization count plus this validation cost. Validation measurements never select the returned vector. The primary comparison uses the exact expected objective at the returned vector; validation means quantify the finite-shot evaluation available to an experimental user. The canonical target is $f_G(\theta) \leq f_G^{\text{ref}} + 0.02$. Hitting cost is the first recorded incumbent’s cumulative optimization shot count satisfying this condition. Runs that never hit the target remain failures, and their costs are retained. The budget-capped target cost assigns the full optimization cap to failures; for successes it uses the first hitting cost. This summary differs from actual incurred shots when a method stops early.

Repeated seeds for one graph do not create additional independent graph instances. The empirical summaries retain the graph as the comparison unit, report depth and budget separately, and distinguish successful hitting costs from success fractions. Classical reference construction and structural preprocessing are reported separately from online simulation time. Simulator timings are not estimates of quantum hardware execution or communication latency. Intervals use the percentile bootstrap [9]: a custom NumPy routine averages shot repetitions within each graph, resamples graph means 10000 times with seed 5713, and takes the 2.5th and 97.5th percentiles of the resampled means. Paired contrasts resample within-graph differences. The follow-up and attribution studies resample within each of their two fixed reference banks. Diagnostic intervals resample whole graphs while retaining all radii, shot counts, depths, shapes, and shot repetitions for a selected graph. Fresh-graph and original-diagnostic intervals are computed separately. Pointwise trajectory bands likewise retain entire graph paths. These intervals describe variability across the observed graphs, not uncertainty over a population of banks. They are descriptive, unadjusted pointwise intervals, without bias correction or acceleration; a band at every radius is not a simultaneous confidence band. Conditional replay bounds instead quantify Monte Carlo error for the fixed proposal cohorts. They do not use 128 replays as 128 new graph instances.

The implementation uses NumPy 2.4.3 for array computation and sampling [17], SciPy 1.17.1 for optimization and linear algebra [36], NetworkX 3.6.1 for graph construction and comparison [16], and Matplotlib 3.10.8 for figures [20]. The environment versions and source hashes saved with the run data identify the implementation used for the reported computations.

The accompanying tests compare the statevector against independently constructed dense Pauli-matrix exponentials, verify the analytic single-edge objective, enumerate cuts independently, and compare weighted QR fits with an SVD-based least-squares solution. Additional checks cover exact graph isomorphism in neighborhood comparison, angle symmetries, coupling marginals and direct four-index distortion, shape determinant and conditioning, budget accounting, and deterministic repeatability. These checks establish specified implementation properties; they do not establish empirical superiority or general optimizer convergence.

4.12 Scientific corrections and experiment provenance

This manuscript replaces the earlier Gaussian-predictor formulation and its empirical aggregates with the independently implemented study described here. Historical aggregate evaluation, timing, calibration, and noise results are not retained because the recovered materials did not include the underlying experiment records needed to reproduce them. The earlier best-of- K derivation also contained an algebraic error. Let $D_k = \|X_k - \mu\|$ for identically distributed draws $X_1, \dots, X_K$ with mean μ and covariance Σ , so that $\mathbb{E}D_k^2 = \text{tr } \Sigma$ for every k . Then $\sqrt{K^{-1} \sum_{k=1}^K \mathbb{E}D_k^2}$ equals $\sqrt{\text{tr } \Sigma}$, not $\sqrt{\text{tr } \Sigma / K}$: averaging K identically distributed squared distances leaves the root-mean-square distance unchanged and supplies no $K^{-1/2}$ factor. No such rate is claimed here. The revised method selects a single donor by its score with a declared numerical tie rule, but no part of the analysis credits that selection with an improvement that grows with the number of bank references. The new empirical record is not a reconstruction of the historical figures.

A Historical controls and supporting evidence

The following results retain the original algorithms and measurements. Their fixed-shot models and contraction after unresolved tests differ from the revised policy. They establish its motivation without being relabelled as revised-policy outcomes.

Exact-data proposals isolate the affine model from sampling error. The offline diagnostic uses 2560 sampled fits across 8 fixed target graphs, both depths, both geometries, five radii and four shot counts. The design is held fixed within each target, depth and geometry. Of the 160 distinct exact-data fits, 84 improve the objective and 79 have a positive sufficient-decrease margin. At 1,024 shots per point, 292/640 sampled proposals improve and 233/640 have a positive sufficient-decrease margin. These dependent proposal counts describe the chosen diagnostic grid, not independent graph observations. Figure 4 separates sampling error around the exact affine fit from total error against the central-finite-difference reference gradient (step 10^{-5}). The reference derivative is evaluated only offline. Neither lower fitting error nor better conditioning alone establishes useful refinement. Generating the model diagnostics uses 83,558,400 simulated observations. The frozen-endpoint replay generates 2,684,354,560 simulated observations, sharing each stored stream across confidence rules and caps. These ideal-sampling offline diagnostic costs are separate from online optimizer cost and do not constitute measurement savings. At radius 0.025, 32/32 exact-data proposals have positive sufficient-decrease margin. At radius 0.4, 0/32 exact-data proposals have positive sufficient-decrease margin.

Frozen endpoint replays measure the cost of decision resolution. Before replay, 160 actual sampled-model proposals are fixed by target, depth, geometry and radius at 1,024 model shots per point, using the first diagnostic replicate. Each condition is replayed 128 times with independent

endpoint streams and common error allocation across confidence rules. The preceding extended replay uses a 131,072-shot endpoint-pair cap and five prescribed looks. Figure 3 now compares all budget prefixes under both four-look and five-look allocations. Acceptance-stage cost includes unresolved runs at their caps. The analytical Hoeffding proposition remains an upper bound for its decision rule, not a universal QAOA shot lower bound. For Hoeffding, the mean acceptance probability over all fixed proposals is 0.000, and the mean unresolved fraction is 0.687. Among the 58 positive-margin proposals, the mean acceptance probability is 0.000, with mean endpoint cost 131,072 shots. For Bernstein, the mean acceptance probability over all fixed proposals is 0.104, and the mean unresolved fraction is 0.446. Among the 58 positive-margin proposals, the mean acceptance probability is 0.288, with mean endpoint cost 128,931 shots.

Paired regime comparisons delimit the evidence. Figure 11 keeps negative effects visible when comparing shaped Bernstein with both conventional optimizers by vertex count and with SPSA by depth within graph families. The two reference banks belong to different target panels, so these comparisons do not identify a bank effect. All intervals average shot repetitions within graph and resample graphs within the two fixed banks (10,000 resamples, seed 5713); they are descriptive and unadjusted. The experiment supplies no fresh holdout validation of a revised policy, large-graph scaling result, structural-shift test, or hardware-noise validation.

A.1 Preserved numerical results

The comparison separates initialization from refinement. The study comprises 4,032 optimization runs on 24 held-out graphs, using two depths, two shot caps, three shot seeds, and 14 methods. All graphs, settings, call records, donor fits, and final validation measurements are saved. The reference bank contains 16 nonisomorphic graphs; held-out graphs are also nonisomorphic to every bank graph. The same graph instances and resource caps are used across all policies in this ideal-circuit simulation study.

Initialization and region shape have distinct effects. At depth 2 and the larger shot cap, descriptor transfer with an isotropic region changes the mean normalized expected cut relative to the fixed-donor control by 1.05 (95% interval 0.65 to 1.51) percentage points, and relative to a random start by 13.95 (95% interval 12.23 to 15.78) percentage points. Keeping the descriptor initialization fixed, the shaped region changes it by 0.00 (95% interval 0.00 to 0.00) percentage points. GW with a shaped region changes it relative to descriptor transfer with an isotropic region by -0.52 (95% interval -0.84 to -0.23) percentage points. Positive values indicate higher expected cut for the method being compared against its stated baseline. Intervals resample the 24 graph-level means after averaging the three shot seeds; they are descriptive, unadjusted percentile bootstrap intervals, not simultaneous confidence statements. The full ablation table prevents an initialization benefit from being attributed to anisotropy (Table 4). The nine undisplaced donor-based local-model policies accept 0 refinement steps in 2,592 runs across both depths and budgets, despite consuming 196,341,504 online shots. For these policies, the final parameter vector equals the transferred anchor: their quality comes entirely from initialization. An explicit initialization-only comparison therefore gives the same depth-2 deficits, 0.38 percentage points for the descriptor anchor and 0.90 for the GW anchor, at zero online refinement shots. This comparison retains the offline donor-construction cost and does not waive any independent output-validation measurements. The 240 accepted moves occur only in the random-start, displaced-prior, and shuffled-reference controls. At the larger depth-2 budget, SPSA has a mean reference deficit of 0.11 percentage points, compared with 0.38 for descriptor-based local-model search and 0.90 for GW-based local-model search. The

prescribed conservative refinement therefore has no demonstrated efficiency advantage over directly using the descriptor-selected anchor. The descriptor initialization already meets the prespecified target on 24/24 graphs at depth 1 and 24/24 graphs at depth 2, before target-graph measurements. The corresponding run counts are 72/72 and 72/72, because each graph contributes three identical initializations; repeated shot seeds add no independent graphs. Zero target-graph shots therefore do not mean zero total cost: donor optimization is performed offline. The target is an expected normalized cut within 0.02 of an independently computed best-found depth-matched reference; it is not a certified QAOA optimum. A negative deficit means that a run exceeds that reference.

Conservative decisions consume the available information. Across all trust-region variants, 240 of 21,484 attempted decisions are accepted and 12,763 remain unresolved at their sampling cap or available budget. Unresolved decisions keep the current point. Figure 8 separates fitting from acceptance measurements. Fixed-shot decisions start at the largest planned sample count; adaptive decisions may stop at an earlier certified look. That single look spends 65,536 shots on the two endpoints, more than the smaller 32,768-shot cap, so the fixed-shot policy can take no acceptance measurement at that cap; 144 unresolved decisions are recorded with zero acceptance shots. Their confidence penalties differ because the fixed-shot variant has one planned look. These are finite-budget comparisons of complete policies, not a demonstration of asymptotic convergence.

An offline evaluation of the recorded endpoints finds 2,898 truly improving proposals among 16,299 trials of the eight adaptive undisplaced donor policies. Of those trials, 15,852 (97.3%) have true acceptance margin $h = f(x) - f(x^+) - \eta q \leq 0.005$; positive decreases are counted above a 10^{-12} numerical tolerance. For this margin range, Proposition 6 bounds the conditional probability of acceptance in any one trial by 3.48×10^{-6} , even if all four prescribed looks are affordable. The fixed-shot control is excluded from this calculation because it uses a different confidence penalty. These exact proposal values are retrospective diagnostics, never online inputs. The counts include dependent trials and are not independent observations or an empirical estimate of the acceptance probability; they connect the transferred-anchor experiment to the regime in which the proposition predicts low acceptance probability.

Building the reference bank used 14,560 exact objective evaluations and 2.3 seconds in the recorded environment. Those exact simulator calls have no assigned hardware-shot equivalent and must not be omitted from deployment considerations. Every final point receives 16,384 additional independent validation shots, one evaluation, and one simulated job. Tables show optimization cost separately so that the common validation cost can be added explicitly. Hardware-job counts describe the specified batching schedule; no device latency or hardware run is inferred. The structural transport comparisons require 9.13 seconds per target graph on average, shared between the two depth-specific banks. Descriptor selection and shape construction require 0.55 milliseconds at depth one. This substantial classical overhead provides an additional reason to prefer the simpler selector for this study; it is not converted into a hardware speedup. Of the 384 distinct target-reference transport solves, 253 end above the prescribed 10^{-9} coupling-change tolerance after the iteration cap. Their couplings satisfy the marginal constraints, but feasibility does not establish convergence or global optimality. These finite-iteration scores must be interpreted together with solver and donor-sensitivity diagnostics rather than as exact graph distances.

Model geometry and the graph bound have limited roles. For descriptor, shaped, the median local-design condition number is 3.99, with 95th percentile 3.99. For no point repair, the median local-design condition number is 17.23, with 95th percentile 223.85. At depth 1, 301/384 target-reference pairs have local-neighborhood total variation equal to one. At depth 2, 384/384

Table 4: Complete depth-2 comparison at an optimization cap of 131,072 shots. Deficit is 100 times the signed difference from the best-found normalized reference objective. Each row averages 24 graphs and three shot seeds. Target success counts all 72 runs, including zero-shot initial successes. Capped target cost assigns the full budget to failures and uses the first successful recorded incumbent for successes. The first two rows are initialization-only policies that return the selected anchor and make no optimization call, so refinement gains are never credited to transfer. Shots exclude 16,384 independent final-validation shots per run, which every policy including initialization alone still incurs. The Local TV and Fixed donor rows differ only through a floating-point tie-break among depth-2 scores that are all equal to one; see the text.

Method	Deficit	Success	Shots	Calls	Jobs	Capped target cost
Descriptor anchor only	0.38	72/72	0	0.0	0.0	0
GW anchor only	0.90	66/72	0	0.0	0.0	10923
Fixed donor	1.43	60/72	130560	64.0	23.0	21845
Random start	14.32	0/72	130400	65.0	23.2	131072
Descriptor, isotropic	0.38	72/72	130560	64.0	23.0	0
Descriptor, shaped	0.38	72/72	130560	64.0	23.0	0
GW, isotropic	0.90	66/72	130560	64.0	23.0	10923
GW, shaped	0.90	66/72	130560	64.0	23.0	10923
Local TV	1.47	60/72	130560	64.0	23.0	21845
Shuffled GW	1.51	54/72	130528	64.2	23.0	32768
Displaced prior	16.91	0/72	130432	65.6	23.4	131072
No point repair	0.38	72/72	130411	65.0	23.2	0
Fixed radius	0.38	72/72	129312	156.7	53.2	0
Fixed shots	0.38	72/72	66816	7.0	2.0	0
COBYLA	0.33	72/72	23282	22.7	22.7	0
SPSA	0.11	72/72	129024	126.0	84.0	0

target-reference pairs have local-neighborhood total variation equal to one. At depth 2 those recorded scores are constant across the bank to within 10^{-15} , which is double-precision rounding in the edge-frequency sums and not structural information. The softmax floor $\tau = 10^{-3}$ in the selection rule nevertheless resolves that rounding, so the local-total-variation anchor differs from the fixed-donor anchor on 3 of 24 depth-2 targets, that is 9 of 72 runs per budget. The separation between the Local TV and Fixed donor rows of Table 4 is an artifact of that tie-break and must not be read as evidence of structural selection. A bound of one cannot narrow the unit range of the normalized objective. Improved numerical conditioning alone is not evidence of saved shots, and a valid transfer bound can be uninformative on the evaluated graphs (Fig. 9). The results do not establish a consistent improvement from the complete transport-shaped policy over the simpler descriptor and conventional-optimizer controls. Comparisons with a tuned ASTRO-DF implementation and physical device noise remain outside this study.

Separately seeded panels test tighter targets and deeper circuits. The separate follow-up contains 448 test runs on 16 graphs from two separately seeded bank-validation-test panels, at depths 2 and 3. Its 128 tuning runs consume 8,170,240 optimization shots and 1,048,576 independent validation shots. The selected model-shot count and initial radius are frozen before any test run and shared by the isotropic and shaped variance-sensitive policies at each depth. All methods use a 65,536-shot online cap; each output receives 8,192 additional validation shots. Table 5 keeps

Table 5: Separate follow-up test comparison. Each deficit is a mean signed reference gap in percentage points with a 95% graph-bootstrap interval, stratified within the two fixed reference banks (10,000 resamples; seed 5713). Each depth averages 16 graph means, with two shot repetitions per graph. Costs are mean online shots; add 8,192 final-validation shots per output and the separately reported tuning and bank costs.

Method	$p = 2$ deficit [95% interval]	$p = 3$ deficit [95% interval]	$p = 2$ shots	$p = 3$ shots
Descriptor anchor	0.41 [0.22, 0.64]	0.77 [0.44, 1.18]	0	0
Fixed schedule	2.94 [2.45, 3.46]	2.74 [2.31, 3.19]	0	0
Hoeffding, isotropic	0.41 [0.22, 0.64]	0.77 [0.44, 1.18]	64,768	64,512
Bernstein, isotropic	0.41 [0.22, 0.64]	0.73 [0.42, 1.14]	64,768	64,512
Bernstein, shaped	0.37 [0.22, 0.56]	0.66 [0.38, 1.01]	64,768	64,512
SPSA	0.21 [0.11, 0.35]	0.41 [0.20, 0.70]	64,512	64,512
COBYLA	0.32 [0.24, 0.41]	0.57 [0.37, 0.82]	22,720	30,752

initialization-only outcomes and actual online costs visible. Exact isomorphism checks identify 1 follow-up test graph also present in the canonical test set, leaving 15 structurally distinct test graphs. There is also 1 bank-to-bank overlap across the two studies. The follow-up bank, validation and test splits remain pairwise nonisomorphic within the new study. At depth 2, validation selects 256 model shots per point and initial radius 0.10. All four validation scores are tied exactly; the first candidate in the declared order wins the tie. This outcome is not evidence of tuning superiority. Hoeffding, isotropic accepts 0 steps and changes 0/32 outputs, using 2,072,576 online shots in total. Bernstein, isotropic accepts 0 steps and changes 0/32 outputs, using 2,072,576 online shots in total. Bernstein, shaped accepts 1 step and changes 1/32 outputs, using 2,072,576 online shots in total. The paired expected-cut change from the isotropic to shaped variance-sensitive rule is 0.03 (95% interval 0.00 to 0.10) percentage points at depth 2; positive values favor shaping. At depth 3, validation selects 256 model shots per point and initial radius 0.10. All four validation scores are tied exactly; the first candidate in the declared order wins the tie. This outcome is not evidence of tuning superiority. Hoeffding, isotropic accepts 0 steps and changes 0/32 outputs, using 2,064,384 online shots in total. Bernstein, isotropic accepts 1 step and changes 1/32 outputs, using 2,064,384 online shots in total. Bernstein, shaped accepts 3 steps and changes 3/32 outputs, using 2,064,384 online shots in total. The paired expected-cut change from the isotropic to shaped variance-sensitive rule is 0.08 (95% interval 0.00 to 0.19) percentage points at depth 3; positive values favor shaping. These graph-level intervals do not estimate uncertainty over a population of reference banks. Separate panel means follow; each panel draws its bank, validation split, test split and random streams together, so a difference between panels is not attributable to the reference bank alone: panel 1, depth 2, gives descriptor-anchor, isotropic Bernstein, shaped Bernstein and SPSA deficits of 0.46, 0.46, 0.46, 0.23 percentage points, respectively. panel 1, depth 3, gives descriptor-anchor, isotropic Bernstein, shaped Bernstein and SPSA deficits of 0.99, 0.92, 0.77, 0.51 percentage points, respectively. panel 2, depth 2, gives descriptor-anchor, isotropic Bernstein, shaped Bernstein and SPSA deficits of 0.35, 0.35, 0.29, 0.19 percentage points, respectively. panel 2, depth 3, gives descriptor-anchor, isotropic Bernstein, shaped Bernstein and SPSA deficits of 0.55, 0.55, 0.55, 0.31 percentage points, respectively. At depth 2, the initialization-only anchor meets gaps 0.002, 0.005, 0.01 and 0.02 on 16/32, 26/32, 28/32, 32/32 test runs, respectively. Repeated initialization-only records correspond to the same anchor within a graph. At depth 3, the initialization-only anchor meets gaps 0.002, 0.005, 0.01 and 0.02 on 6/32, 18/32, 24/32, 28/32 test runs, respectively. Repeated

Table 6: Transport sensitivity on 384 canonical graph pairs. “Unconv.” and “Infeas.” count selected pairwise solver candidates; changed donors and valid targets count graphs out of 24. Depth-specific deficits are mean anchor-only signed reference gaps in percentage points over valid targets. The first two rows are reference points rather than audit solvers: the study’s own capped entropic solver and the transport-free descriptor selector, both over all 24 targets. Feasibility, convergence, and useful donor selection are distinct properties.

Solver	Unconv.	Infeas.	Changed donors	Valid targets	$p = 1$ deficit	$p = 2$ deficit
Original capped entropic (study)	253	0	–	24	0.29	0.90
Descriptor selector (no transport)	–	–	–	24	0.09	0.38
Entropic $\epsilon = 0.02$	100	99	1	1	0.15	0.50
Entropic $\epsilon = 0.05$	3	0	1	24	0.30	0.89
Entropic $\epsilon = 0.1$	5	0	20	24	0.31	1.01
Entropic, three starts	14	0	10	24	0.23	0.74
Conditional gradient, three starts	0	0	21	24	0.24	0.65

initialization-only records correspond to the same anchor within a graph. An exclusion sensitivity analysis removes the overlapping graph without refitting or rerunning any method. On these 15 graphs at depth 2, the paired expected-cut change from isotropic to shaped Bernstein is 0.03 percentage points; the lowest mean deficits are SPSA (0.22) and COBYLA (0.33). The shaped rule has deficit 0.39, compared with 0.43 for initialization alone. On these 15 graphs at depth 3, the paired expected-cut change from isotropic to shaped Bernstein is 0.08 percentage points; the lowest mean deficits are SPSA (0.43) and COBYLA (0.58). The shaped rule has deficit 0.67, compared with 0.78 for initialization alone. The held-out test runs consume 18,250,752 optimization shots and 3,670,016 additional validation shots. All follow-up best-found references together use 97,183 exact objective calls, including 40,409 bank-reference calls; these are classical simulation costs, not uncharged hardware shots. The added panels enlarge the tested regime but remain ideal-circuit, small-graph experiments with a fixed, disclosed tuning grid.

Solver sensitivity limits structural-selection conclusions. The separate transport audit repeats every canonical target–reference comparison with longer iterations, three regularizations, multiple starts, and independent conditional-gradient optimization. All saved couplings are checked by a direct four-index distortion calculation. Conditional-gradient stopping is checked by a transportation linear-program gap; this is a first-order stationarity diagnostic at the stated numerical tolerance, not a certificate of exact stationarity or global optimality. Table 6 reports solver failures alongside changed donors and resulting anchor quality. A target with any infeasible candidate is excluded from that selector’s quality summary and explicitly counted; no finite score is substituted for an invalid solve. Quality means over different valid-target subsets are not paired policy comparisons and must not be used to rank those selectors directly.

A.2 Descriptive effects across existing sizes and depths

These comparisons concern sizes 12 and 14 and depths two and three on the existing target set. They are conditional descriptions, not asymptotic scaling laws or bank-robustness tests. Each bank serves different targets, so panel differences cannot isolate a bank effect.

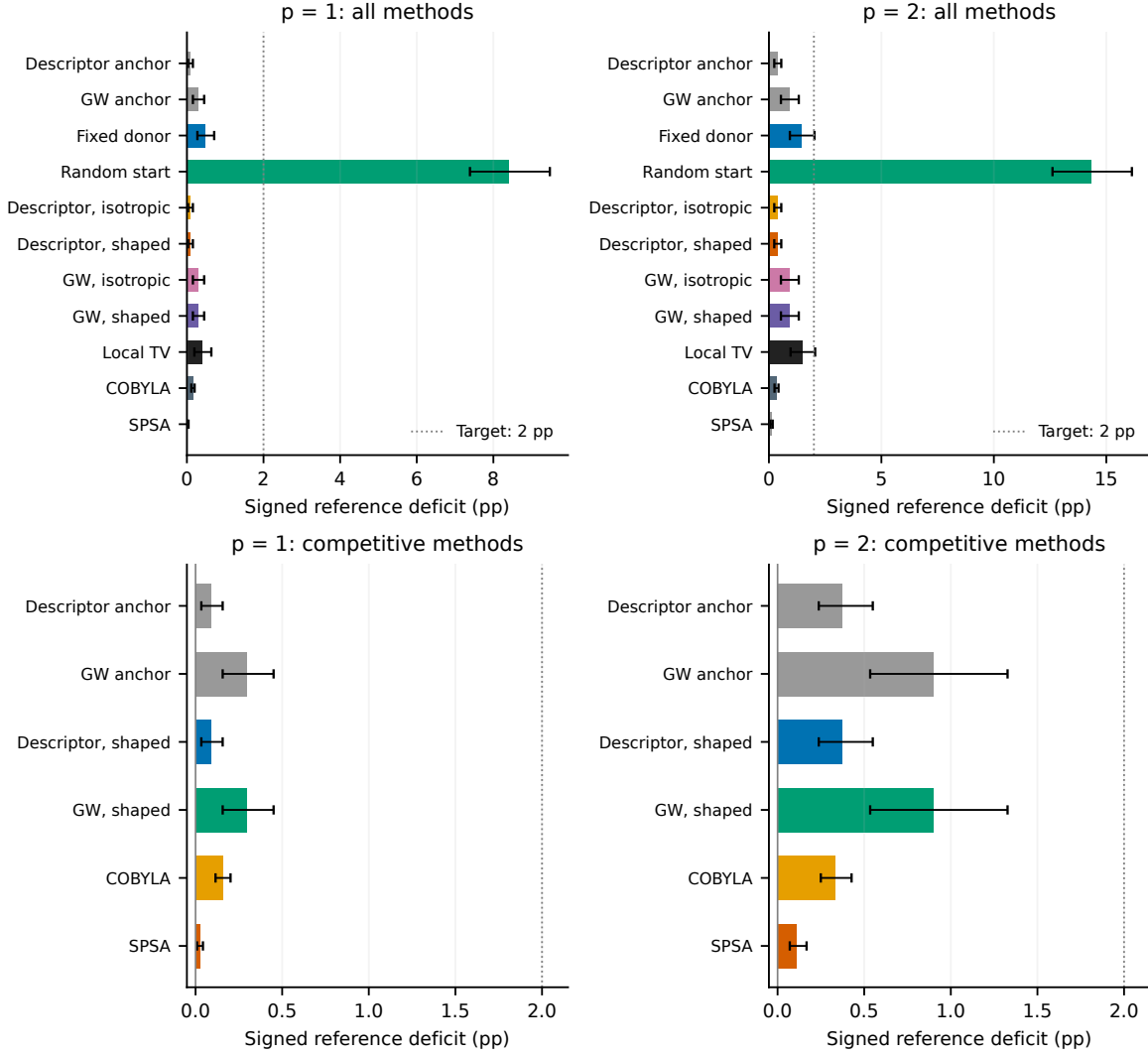


Figure 6: Final quality at a 131072-shot optimization cap, at depths one and two. Deficit is 100 times the signed gap between the returned normalized objective and the best-found reference; smaller is better. The dotted line marks the target gap of 0.02, or two percentage points on the plotted scale. Separate panels retain all methods and show the competitive methods from a zero origin; initialization-only controls expose the transferred quality. Bars average the 24 graph means after averaging three shot seeds within each graph. Intervals use 10000 graph-bootstrap resamples (95% percentile). Exact objectives are diagnostic; independent final validation is outside the cap.

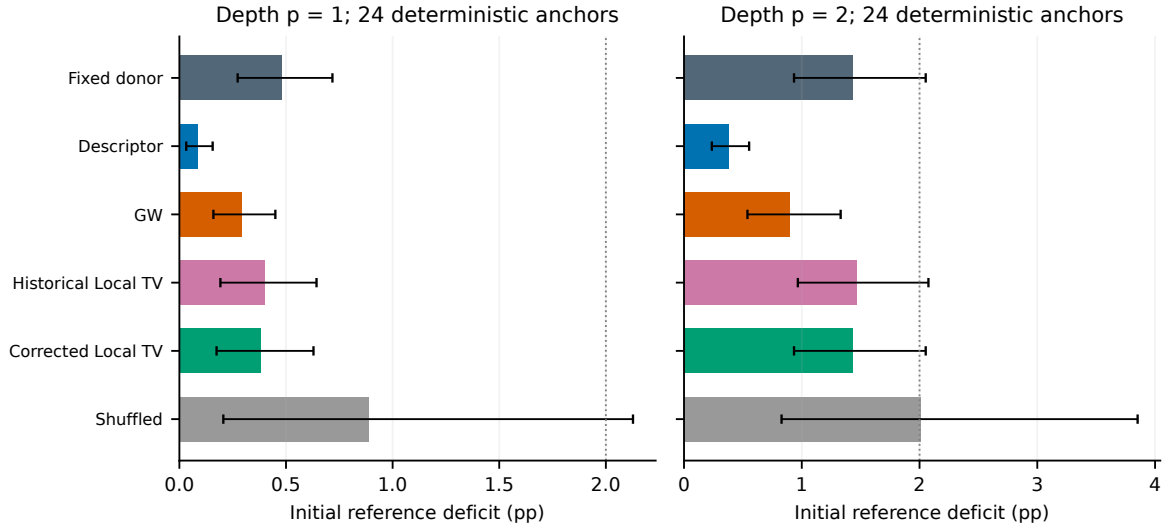


Figure 7: Initial transfer deficits, in percentage points, before target-graph refinement. Bars summarize deterministic initial deficits on 24 graphs; intervals use 10000 graph-bootstrap resamples (95% percentile). Horizontal scales differ between depths. Best-found target references are used only for evaluation. Theorem 1 applies to rooted-neighborhood total variation; the other selectors are heuristic. All depth-two local scores mathematically equal one: the historical Local-TV row retains the floating-point selection artifact, while the corrected deterministic-tie row selects the fixed donor. This separation is not evidence of graph structure.

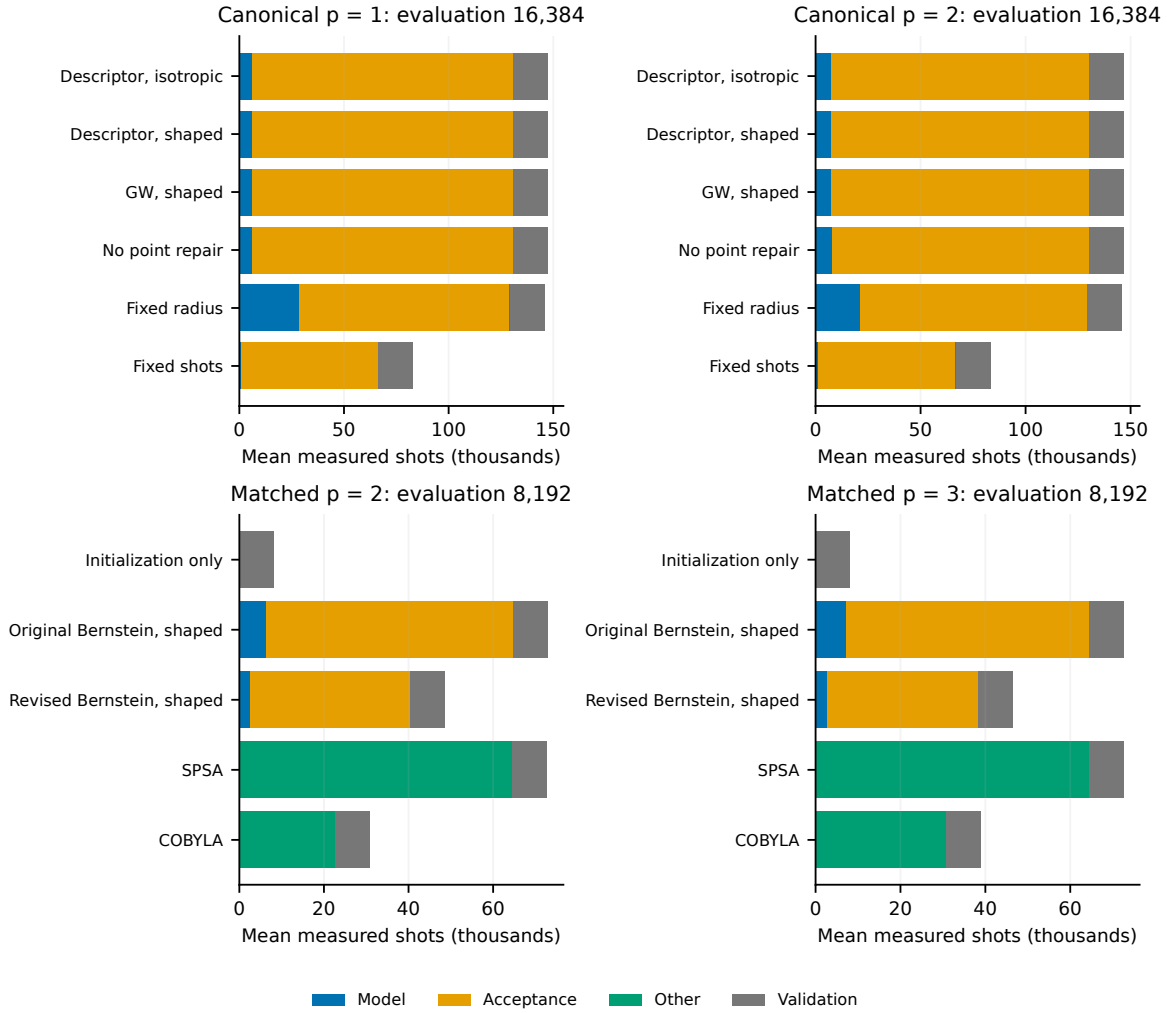


Figure 8: Measurement expenditure across the canonical and matched studies. Stacked bars separate model, acceptance, and independent final-evaluation shots. Canonical panels use a 131072-shot optimization cap at depths one and two; matched follow-up and revision panels use a 65536-shot cap at depths two and three. Rejected and unresolved trials remain charged. Independent evaluation costs 16384 shots per canonical run and 8192 per matched run. These bars report expenditure rather than optimization trajectories.

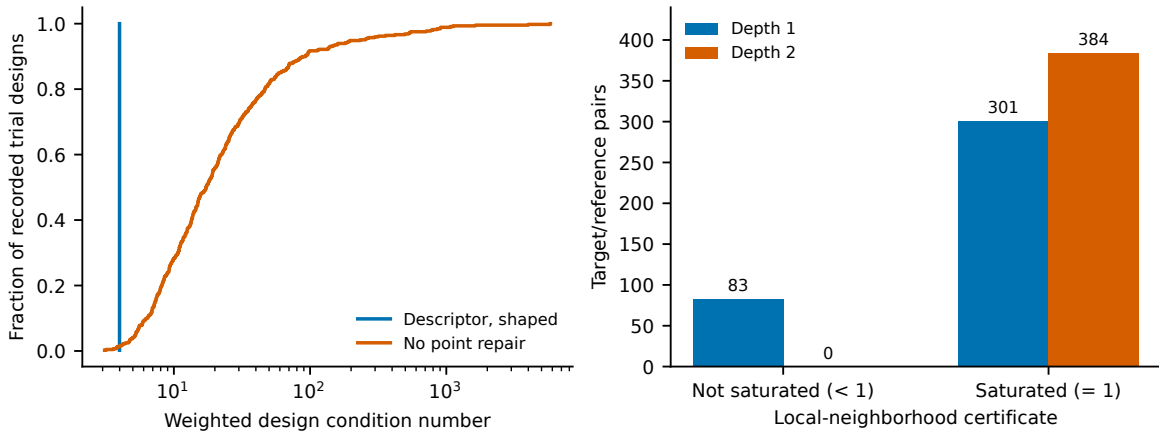


Figure 9: Interpolation geometry and certificate saturation. Step-line empirical cumulative distributions compare recorded weighted-design condition numbers. Grouped bars count not-saturated and saturated rooted neighborhood bounds: 83 versus 301 pairs at depth one and 0 versus 384 at depth two. Saturation means total variation one and a vacuous objective bound; a value below one need not be practically useful. The ECDF weights each recorded model equally, so it describes the executed trial mixture rather than giving equal weight to graphs. Good numerical conditioning alone does not establish refinement.

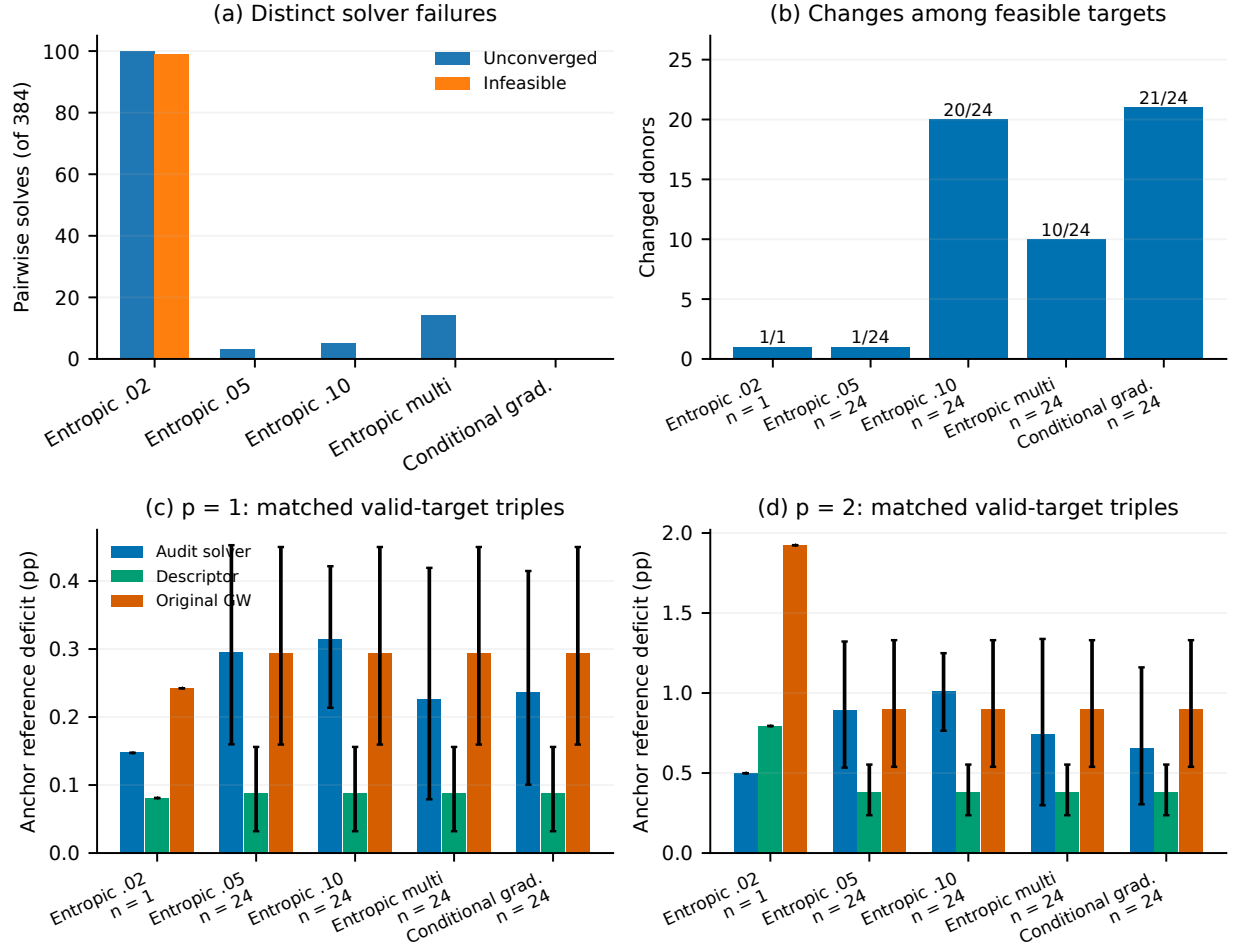


Figure 10: Transport-selector audit. Separate bar panels report solver feasibility and changed donors with explicit target denominators. Quality comparisons use common feasible target sets and include descriptor and original-solver references. The single aggregate audit time is reported in text because per-solver timings are unavailable. The descriptor comparator supplies the inexpensive default used by the revised refinement experiment.

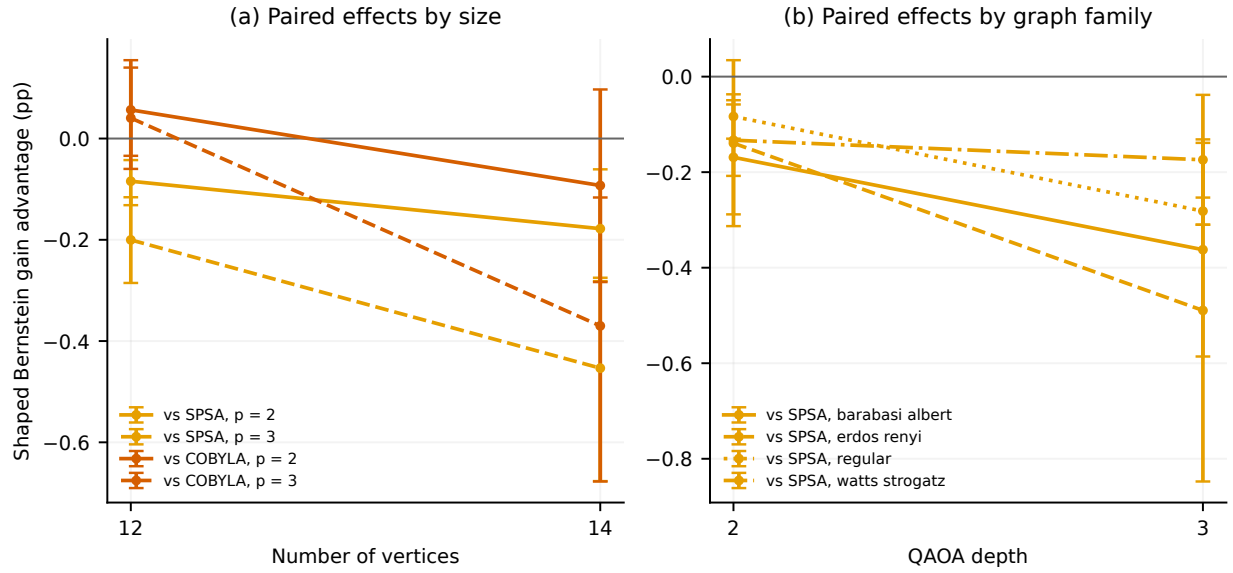


Figure 11: Paired performance across the tested regimes. Lines show the revised policy's gain difference from SPSA and COBYLA across graph sizes (left), and from SPSA across depths within each graph family (right). Zero marks equal gain, and negative effects remain visible. Repetitions are averaged within graphs before uncertainty estimation. Comparisons are conditional on the two existing banks; changing targets across panels prevents an isolated estimate of a bank effect.

References

- [1] Albert-László Barabási and Réka Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999. doi: 10.1126/science.286.5439.509.
- [2] Fernando G. S. L. Brandão, Michael Broughton, Edward Farhi, Sam Gutmann, and Hartmut Neven. For fixed control parameters the quantum approximate optimization algorithm’s objective function value concentrates for typical instances, 2018. URL <https://arxiv.org/abs/1812.04170>. arXiv:1812.04170.
- [3] Peter Businger and Gene H. Golub. Linear least squares solutions by Householder transformations. *Numerische Mathematik*, 7:269–276, 1965. doi: 10.1007/BF01436084.
- [4] Ruobing Chen, Matt Menickelly, and Katya Scheinberg. Stochastic optimization using a trust-region method and random models. *Mathematical Programming*, 169:447–487, 2018. doi: 10.1007/s10107-017-1141-8. arXiv:1504.04231.
- [5] Andrew R. Conn, Katya Scheinberg, and Luís N. Vicente. Geometry of interpolation sets in derivative free optimization. *Mathematical Programming*, 111:141–172, 2008. doi: 10.1007/s10107-006-0073-5.
- [6] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, 2013. URL https://papers.neurips.cc/paper_files/paper/2013/hash/af21d0c97db2e27e13572cbf59eb343d-Abstract.html.
- [7] Pinar Demetci, Rebecca Santorella, Björn Sandstede, William Stafford Noble, and Ritambhara Singh. SCOT: Single-cell multi-omics alignment with optimal transport. *Journal of Computational Biology*, 29(1):3–18, 2022. doi: 10.1089/cmb.2021.0446.
- [8] Kwassi Joseph Dzahini, Jeffrey M. Larson, Matt Menickelly, and Stefan M. Wild. A noise-aware scalable subspace classical optimizer for the quantum approximate optimization algorithm. In *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 20–30. IEEE, 2025. doi: 10.1109/QCE65121.2025.00013. URL <https://doi.org/10.1109/QCE65121.2025.00013>.
- [9] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*, volume 57 of *Monographs on Statistics and Applied Probability*. Chapman & Hall, New York, 1993. ISBN 0-412-04231-2.
- [10] Jose Falla, Quinn Langfitt, Yuri Alexeev, and Ilya Safro. Graph representation learning for parameter transferability in quantum approximate optimization algorithm. *Quantum Machine Intelligence*, 6:46, 2024. doi: 10.1007/s42484-024-00178-9.
- [11] Edward Farhi, Jeffrey Goldstone, and Sam Gutmann. A quantum approximate optimization algorithm, 2014. URL <https://arxiv.org/abs/1411.4028>. arXiv:1411.4028.
- [12] Alexey Galda, Eesh Gupta, Jose Falla, Xiaoyuan Liu, Danylo Lykov, Yuri Alexeev, and Ilya Safro. Similarity-based parameter transferability in the quantum approximate optimization algorithm. *Frontiers in Quantum Science and Technology*, 2:1200975, 2023. doi: 10.3389/frqst.2023.1200975. arXiv:2307.05420.
- [13] E. N. Gilbert. Random graphs. *The Annals of Mathematical Statistics*, 30(4):1141–1144, 1959. doi: 10.1214/aoms/1177706098.

- [14] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, Baltimore, 4 edition, 2013. doi: 10.56021/9781421407944.
- [15] Yunsoo Ha, Sara Shashaani, and Matt Menickelly. Two-stage estimation and variance modeling for latency-constrained variational quantum algorithms. *INFORMS Journal on Computing*, 37(1):125–145, 2025. doi: 10.1287/ijoc.2024.0575. URL <https://pubsonline.informs.org/doi/10.1287/ijoc.2024.0575>.
- [16] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference*, pages 11–15, Pasadena, CA, 2008. URL <https://aric.hagberg.org/papers/hagberg-2008-exploring.pdf>.
- [17] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.
- [18] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.
- [19] Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, non-parametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021. doi: 10.1214/20-AOS1991. URL <https://arxiv.org/abs/1810.08240>.
- [20] J. D. Hunter. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007. doi: 10.1109/MCSE.2007.55.
- [21] Jonas M. Kübler, Andrew Arrasmith, Lukasz Cincio, and Patrick J. Coles. An adaptive optimizer for measurement-frugal variational algorithms. *Quantum*, 4:263, 2020. doi: 10.22331/q-2020-05-11-263. URL <https://doi.org/10.22331/q-2020-05-11-263>.
- [22] Simon Lacoste-Julien. Convergence rate of Frank–Wolfe for non-convex objectives, 2016. URL <https://arxiv.org/abs/1607.00345>. arXiv:1607.00345.
- [23] Andreas Maurer and Massimiliano Pontil. Empirical Bernstein bounds and sample variance penalization. In *Proceedings of the 22nd Annual Conference on Learning Theory*, 2009. doi: 10.48550/arXiv.0907.3740. URL <https://www.cs.mcgill.ca/~colt2009/papers/012.pdf>. arXiv:0907.3740.
- [24] Facundo Mémoli. Gromov–Wasserstein distances and the metric approach to object matching. *Foundations of Computational Mathematics*, 11(4):417–487, 2011. doi: 10.1007/s10208-011-9093-5.
- [25] Kien X. Nguyen and Ilya Safro. Graph-conditioned meta-optimizer for QAOA parameter generation on multiple problem classes, 2026. URL <https://arxiv.org/abs/2604.25275>. arXiv:2604.25275.

- [26] Jorge Nocedal and Stephen J. Wright. *Numerical Optimization*. Springer Series in Operations Research and Financial Engineering. Springer, New York, 2 edition, 2006. doi: 10.1007/978-0-387-40065-5.
- [27] Shubham Patel and Utkarsh Mishra. Improving the efficiency of QAOA using efficient parameter transfer initialization and targeted-single-layer regularized optimization with minimal performance degradation, 2026. URL <https://arxiv.org/abs/2601.15760>. arXiv:2601.15760.
- [28] Gabriel Peyré, Marco Cuturi, and Justin Solomon. Gromov–Wasserstein averaging of kernel and distance matrices. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 2664–2672, 2016. URL <https://proceedings.mlr.press/v48/peyre16.html>.
- [29] M. J. D. Powell. A direct search optimization method that models the objective and constraint functions by linear interpolation. In Susana Gomez and Jean-Pierre Hennart, editors, *Advances in Optimization and Numerical Analysis*, volume 275 of *Mathematics and Its Applications*, pages 51–67. Springer, Dordrecht, 1994. doi: 10.1007/978-94-015-8330-5_4.
- [30] Stefan H. Sack and Maksym Serbyn. Quantum annealing initialization of the quantum approximate optimization algorithm. *Quantum*, 5:491, 2021. doi: 10.22331/q-2021-07-01-491. URL <https://doi.org/10.22331/q-2021-07-01-491>.
- [31] Sara Shashaani, Fatemeh S. Hashemi, and Raghu Pasupathy. ASTRO-DF: A class of adaptive sampling trust-region algorithms for derivative-free stochastic optimization. *SIAM Journal on Optimization*, 28(4):3145–3176, 2018. doi: 10.1137/15M1042425.
- [32] James C. Spall. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE Transactions on Automatic Control*, 37(3):332–341, 1992. doi: 10.1109/9.119632.
- [33] A. Steger and N. C. Wormald. Generating random regular graphs quickly. *Combinatorics, Probability and Computing*, 8(4):377–396, 1999. doi: 10.1017/S0963548399003867.
- [34] Michael Streif and Martin Leib. Training the quantum approximate optimization algorithm without access to a quantum processing unit. *Quantum Science and Technology*, 5(3):034008, 2020. doi: 10.1088/2058-9565/ab8c2b. URL <https://doi.org/10.1088/2058-9565/ab8c2b>.
- [35] Kevin J. Sung, Jiahao Yao, Matthew P. Harrigan, Nicholas C. Rubin, Zhang Jiang, Lin Lin, Ryan Babbush, and Jarrod R. McClean. Using models to improve optimizers for variational quantum algorithms. *Quantum Science and Technology*, 5(4):044008, 2020. doi: 10.1088/2058-9565/abb6d9. URL <https://doi.org/10.1088/2058-9565/abb6d9>.
- [36] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.

- [37] Duncan J. Watts and Steven H. Strogatz. Collective dynamics of 'small-world' networks. *Nature*, 393:440–442, 1998. doi: 10.1038/30918.
- [38] Jonathan Wurtz and Danylo Lykov. Fixed-angle conjectures for the quantum approximate optimization algorithm on regular MaxCut graphs. *Physical Review A*, 104(5):052419, 2021. doi: 10.1103/PhysRevA.104.052419. URL <https://doi.org/10.1103/PhysRevA.104.052419>.
- [39] Zaikun Zhang. PRIMA: Reference implementation for Powell’s methods with modernization and amelioration, 2023. URL <https://www.libprima.net>.
- [40] Leo Zhou, Sheng-Tao Wang, Soonwon Choi, Hannes Pichler, and Mikhail D. Lukin. Quantum approximate optimization algorithm: Performance, mechanism, and implementation on near-term devices. *Physical Review X*, 10:021067, 2020. doi: 10.1103/PhysRevX.10.021067.